INDIAN INSTITUTE OF TECHNOLOGY MADRAS

System Commands (Linux / Bash)

Past Year Question Bank (33 Exam Sessions) — Official Categorized Syllabus Edition

Total Unique Questions: 189	**Exam Sessions Covered:** 33 Papers	**Format:** MCQ, MSQ, NAT / SA
Week 1: Shell & Basics (75 Qs)	**Week 2:** Permissions & Streams (21 Qs)	**Week 3:** Filters, Grep & Awk (53 Qs)
Week 4: Scripting & Processes (40 Qs)	**Fidelity:** Official TCS iON Cards	**Interactive Mode:** NTA JEE CBT Online

Week 1: Introduction to Operating Systems, Shell & Basic Commands (75 Questions)

- **1.2 Directory Navigation & Pathnames** **(8 Questions)**

SC_001 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ EXAM QPG1 (+1 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.2 Directory Navigation & Pathnames

Question Number : 152 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What would be the output of the last command after executing the following command sequence?

```
$ pwd
/home/ubuntu
$ cd dir1
$ cd .
$ cd dir2
$ cd ..
$ pwd
```

Options :

A. ✔ `/home/ubuntu/dir1`

B. ✖ `/home/ubuntu/dir2`

C. ✖ `/home/ubuntu/dir1/dir2`

D. ✖ `/home/ubuntu`

✔ Official Answer Key: Option (A), Option (B), Option (D)

SC_013 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.2 Directory Navigation & Pathnames

Question Number : 266 Question Id : 640653345412 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

```
ls | grep "^[^a-z]"
```

The above command will print the filenames in the current working directory that,

Options :

6406531148818. ✖ Starts with an uppercase letter.

6406531148819. ✖ Starts with a lowercase letter.

6406531148820. ✖ Does not start with an uppercase letter.

6406531148821. ✔ Does not start with a lowercase letter.

✔ Official Answer Key: Option (D)

SC_020 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.2 Directory Navigation & Pathnames

Question Number : 273 Question Id : 640653345420 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Select Question

Select the command(s) that prints the file/directory names, that start with the letter "a" and end with the letter "e" (case-sensitive) present in the current working directory.

Options :

6406531148846. ✔ `ls | grep '^a.*e$'`

6406531148847. ✖ `ls | grep 'a.*e'`

6406531148848. ✔ `echo a*e`

6406531148849. ✖ `echo *a*e*`

✔ Official Answer Key: Option (A), Option (C)

SC_021 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.2 Directory Navigation & Pathnames

Question Number : 274 Question Id : 640653345424 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Select Question

Consider the long listing output of the present working directory below.

```
$ ls -l
total 5
-rw-r--r-- 1 ram      teacher          315 May 17 17:18 Quiz1.md
-rw-r--r-- 1 megha    studentsDiploma   10 May 17 17:18 assignment1.txt
-rw-r--rw- 1 ravi     teacher           24 May 17 17:07 assignment2.txt
drwxr-xr-x 1 guest    guest              0 Dec 20 09:37 abc.ab
```

Which of the following statement(s) are true?

Options :

6406531148864. ✔ The owner of the file `Quiz1.md` is `ram`.

6406531148865. ✖ The owner of the file `Quiz1.md` is `teacher`.

6406531148866. ✔ `ram` does not have write permissions on the file `assignment1.txt`.

6406531148867. ✖ `ram` does not have read permissions on the file `assignment1.txt`.

6406531148868. ✖ If `ram` is in the group named `teacher`, then he has execute permission on the file `assignment2.txt`.

6406531148869. ✖ If `ram` is in the group named `teacher`, then he has write permission on the file `assignment2.txt`.

✔ Official Answer Key: Option (A), Option (C)

SC_025 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.2 Directory Navigation & Pathnames

Question Number : 278 Question Id : 640653345419 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Select Question

Select the command(s) that will return with an error, given that only file `a` is present initially in the current working directory.

Options :

6406531148840. ✖ `cp a b # Command 1`

6406531148841. ✖ `mv b c # Command 2`

6406531148842. ✖ `mv c d # Command 3`

6406531148843. ✖ `mv a d # Command 4`

6406531148844. ✔ `cp a b # Command 5`

6406531148845. ✔ `mv b d # Command 6`

✔ Official Answer Key: Option (E), Option (F)

SC_070 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.2 Directory Navigation & Pathnames

Question Number : 253 Question Id : 640653737537 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Assume that each directory has 2 hard links (one for the current working directory and another from its parent directory) and a link each for the sub-directory's parent directory. How many subdirectories are present in the current working directory?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

14

✔ **Official Answer Key (Numerical Value): 14**

SC_080 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.2 Directory Navigation & Pathnames

Question Number : 248 Question Id : 6406531113966 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following JSON file `people.json` which is present in the current working directory.

```
{
    "people": [
        {
            "firstName": "Alice",
            "sequence": 26803
        },
        {
            "firstName": "Bob",
            "sequence": 8629
        },
        {
            "firstName": "Charlie",
            "sequence": 27799
        },
        {
            "firstName": "Dan",
            "sequence": 4749
        }

    ]
}
```

Which of the following commands can be used to extract the highest sequence number?

Options :

6406533774718. ✔ `cat people.json | egrep -o [[:digit:]]+ | sort -nr | head -n 1`

6406533774719. ✖ `cat people.json | egrep -o [[:digit:]]+ | sort -n | head -n 1`

6406533774720. ✔ `cat people.json | egrep -o [[:digit:]]+ | sort -n | tail -n 1`

6406533774721. ✖ `cat people.json | egrep -o [[:digit:]]+ | sort -nr | tail -n 1`

✔ Official Answer Key: Option (A), Option (C)

SC_155 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.2 Directory Navigation & Pathnames

Question Number : 235 Question Id : 640653668678 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7

Question Label : Short Answer Question

The following is an output of the 'ls -li' command. Based on the information present in the output; how many hardlinks are present for the parent directory (`testdir`).

Hint: Each directory has two hardlinks, one for `.` and another for `..` and in addition to it, each subdirectory increments the count of hardlinks to the parent directory by one.

```
$ cd testdir
$ ls -li
total 68
122703 drwxr-xr-x 2 systemcommands systemcommands  4096 Sep 25 11:28 2021Aug
116829 drwxr-xr-x 2 systemcommands systemcommands  4096 Sep 25 06:49 folder1
116830 drwxr-xr-x 2 systemcommands systemcommands  4096 Sep 25 06:49 folder2
116831 drwxr-xr-x 2 systemcommands systemcommands  4096 Sep 25 07:10 folder3
122704 -rw-r--r-- 1 systemcommands systemcommands  8228 Sep 25 09:37
html.txt
122705 -rw-r--r-- 1 systemcommands systemcommands   605 Sep 25 09:53
onecolor.txt
122706 -rw-r--r-- 1 systemcommands systemcommands   440 Sep 25 10:38
onecolor1.txt
 82472 drwxr-xr-x 2 systemcommands systemcommands 12288 Sep 25 08:49
test_directory
116840 -rwxr-xr-x 1 systemcommands systemcommands   225 Sep 25 08:42
testdata.sh
116857 -rwxr-xr-x 1 systemcommands systemcommands   351 Sep 25 08:48
testdata2.sh
```

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

7

✔ Official Answer Key (Numerical Value): 7

• 1.1 OS Architecture, Linux Kernel & Shell Basics (47 Questions)

SC_002 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ EXAM QPG1 (+1 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 157 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following commands will display the files whose name does not start with a lower case alphabet?

Options :

A. ✔ `ls | egrep '^[^a-z]'`

B. ✖ `ls | egrep '^^[a-z]'`

C. ✖ `ls | egrep '^[a-z]'`

D. ✖ `ls | egrep '[^a-z]'`

✔ Official Answer Key: Option (A)

SC_003 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ EXAM QPG1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 169 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

How many directories will be created by the command

`mkdir dir-{0..10} && mkdir dir-100 || mkdir dir-101` ? Enter the number.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12

Question Type : COMPREHENSION

Question Numbers : (170 to 171)

Question Label : Comprehension

There are four files `log1` , `log2` , `log11` and `log22` .

- `log2` is a hard link to `log1`
- `log11` is a soft link to `log1`
- And `log22` is soft link to `log2`

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 12**

SC_004 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ EXAM QPG1 (+1 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 170 Question Type : MSQ

Correct Marks : 2

Question Label : Multiple Select Question

Which of the following statements are true?

Options :

A. ✖ Inode of `log1` and `log11` will be the same.

B. ✔ Inode of `log1` and `log2` will be the same.

C. ✖ Inode of `log2` and `log22` will be the same.

D. ✖ Inode of `log11` and `log22` will be the same.

✔ Official Answer Key: Option (B)

SC_005 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ EXAM QPG2*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 169 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

How many directories will be created by the command

`mkdir dir-{0..9} && mkdir dir-100 || mkdir dir-101` ? Enter the number.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

11

Question Type : COMPREHENSION

Question Numbers : (170 to 171)

Question Label : Comprehension

There are four files `log1` , `log2` , `log11` and `log22` .

- `log1` is a hard link to `log2`
- `log11` is a soft link to `log1`
- And `log22` is soft link to `log2`

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key (Numerical Value): 11**

SC_008 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 261 Question Id : 640653345416 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Select Question

```
echo 'Hello, World' >out 1>&2
```

Which of the following statement(s) will be true after executing the above command.

Options :

6406531148835. ✖ The file `out` contains the text `Hello, World`.

6406531148836. ✔ The text `Hello, World` will be printed on the terminal.

6406531148837. ✔ The file `out` contains no text (i.e. will be an empty file).

6406531148838. ✖ No text will be printed on the terminal.

Sub-Section Number :	3
Sub-Section Id :	64065348826
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B), Option (C)

SC_014 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 267 Question Id : 640653345415 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

```
$ ps
    PID TTY          TIME CMD
  64754 pts/3    00:00:00 bash
  65657 pts/3    00:00:00 sleep
  65671 pts/3    00:00:00 sleep
  65718 pts/3    00:00:00 python3
  65774 pts/3    00:00:00 ps

$ ps | grep "$$"
```

What will be the output of the last command in the above sequence of commands?

Options :

6406531148831. ✖ `PID TTY TIME CMD`

6406531148832. ✔ `64754 pts/3 00:00:00 bash`

6406531148833. ✖ `65718 pts/3 00:00:00 python3`

6406531148834. ✖ `65774 pts/3 00:00:00 ps`

✔ Official Answer Key: Option (B)

SC_019 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 272 Question Id : 640653345409 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Below are the commands executed in the given sequence in an empty directory(containing no files or directories). Enter the number of files that will be created at the end of the execution of the below commands.

```
# Format Strings
# %H     hour (00..23)
# %M     minute (00..59)

fileName=$(date +'%M')
touch $fileName; sleep 3600; touch $fileName; sleep 3600
```

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Sub-Section Number :	8
Sub-Section Id :	64065348831
Question Shuffling Allowed :	Yes

✔ Official Answer Key (Numerical Value): 1

SC_022 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 275 Question Id : 640653345425 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Select Question

Consider the contents of the file `employee_details.txt` below. The employees are ordered in the increasing order of their age in the file from top to bottom.

```
$ cat employee_details.txt
A1998001,Ram Kumar,7,Male
B2000002,Sanjay Narayan,10,Male
B2000003,Srishti Rai,12,Female
E1997001,Manoj Pillai,14,Male
G1998001,Preeti Suresh,18,Female
G1999001,Leela L G,21,Female
```

Which of the following command will output the full names of the 3 youngest employees, one name on each line.

Options :

6406531148870. ✔ `cat employee_details.txt | head -3 | cut -d "," -f 2`

6406531148871. ✔ `cat employee_details.txt | cut -d "," -f 2 | head -3`

6406531148872. ✖ `cat employee_details.txt | tail -3 | cut -d "," -f 2`

6406531148873. ✖ `cat employee_details.txt | head -3 | cut -f 3`

6406531148874. ✖ `cat employee_details.txt | cut -d "," -f 3 | tail -3`

✔ Official Answer Key: Option (A), Option (B)

SC_024 | [SA] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 277 Question Id : 640653345418 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Initially, file `a` exists in the current working directory. How many files will be present in the current working directory after executing the given commands in the same sequence?

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

✔ **Official Answer Key (Numerical Value): 1**

SC_028 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 229 Question Id : 6406531298623 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following table of network ports and services:

Port	Service
1. 22	A. http (Hypertext Transfer Protocol)
2. 80	B. smtp (Simple Mail Transfer Protocol)
3. 25	C. ssh (Secure Shell)
4. 443	D. https (Secure Hypertext Transfer Protocol)

Which of the following options correctly matches all the ports to their respective services as listed in the provided information?

Options :

6406534366691. ✖ 1-A, 2-B, 3-C, 4-D

6406534366692. ✔ 1-C, 2-A, 3-B, 4-D

6406534366693. ✖ 1-B, 2-D, 3-A, 4-C

6406534366694. ✖ 1-D, 2-C, 3-A, 4-B

✔ Official Answer Key: Option (B)

SC_029 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 230 Question Id : 6406531298624 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

You are trying to connect to a remote server using its IP address with the command `ssh user@192.168.1.50`. However, the command takes a very long time and eventually fails, showing a "Connection timed out" error.

Based on the information about network connectivity and tools, what is a simple, initial step you could take to check a very basic reason for this failure?

Options :

6406534366695. ✖ Try using a different command-line editor like `vi` or `emacs` to launch the SSH command, as your current editor might be incompatible.

6406534366696. ✔ Check if the remote server at `192.168.1.50` is powered on and reachable on the network using the `ping 192.168.1.50` command.

6406534366697. ✖ Assume the SSH service (sshd) on the server has crashed and immediately try to restart it using a web-based control panel, if available.

6406534366698. ✖ The server is likely using a non-standard port for SSH, so you should try connecting to common web ports like 80 or 443 instead.

✔ Official Answer Key: Option (B)

SC_031 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 232 Question Id : 6406531298626 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The primary function of the pipe operator (`|`) as in `command1 | command2` is to:

Options :

6406534366703. ✖ Run `command1` and `command2` in sequence, regardless of success.

6406534366704. ✖ Send the `stderr` of `command1` to the `stdin` of `command2`.

6406534366705. ✖ Execute `command2` only if `command1` outputs data.

6406534366706. ✔ Send the `stdout` of `command1` to the `stdin` of `command2`.

✔ Official Answer Key: Option (D)

SC_042 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 243 Question Id : 640653470180 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

```
ALPHA="A"
GAMMA="G"
echo "$ALPHA$BETA$GAMMA"
```

Select the output from the above script.

Options :

6406531562453. ✖ ALPHABETAGAMMA

6406531562454. ✖ ABG

6406531562455. ✖ ABETAG

6406531562456. ✔ AG

✔ Official Answer Key: Option (D)

SC_046 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 247 Question Id : 640653470181 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Short Answer Question

```
a=12
b=56
c=$(a + b)
d=$(a + c)
```

What will be the value of `d` at the end of execution?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

80

✔ **Official Answer Key (Numerical Value): 80**

SC_054 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 255 Question Id : 640653470191 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 5

Question Label : Short Answer Question

Enter the number `A` that will print the output "SUCCESS"

```
[ "$A" -gt 10 ] &&
[ "$A" -lt 15 ] &&
[ "$(( A % 3 ))" -eq 0 ] &&
echo SUCCESS
```

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12

Sub-Section Number :	7
Sub-Section Id :	64065367734
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (Numerical Value): 12**

SC_061 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 244 Question Id : 640653737533 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following command(s) will run the `ls` command with the `more` command to show the output page-by-page.

Options :

6406532468354. ✔ `ls | more`

6406532468355. ✖ `list="ls | more"; echo $list`

6406532468356. ✖ `list="ls | more"; bash $list`

6406532468357. ✔ `list="ls | more"; eval $list`

✔ Official Answer Key: Option (A), Option (D)

SC_066 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 249 Question Id : 640653737531 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

```
$ start=5; end=8; echo {$start..$end}
```

Above commands were executed in Bash shell. Which of the following represents the output of the above commands.

Options :

6406532468346. ✔ `{5..8}`

6406532468347. ✖ `5 6 7 8`

6406532468348. ✖

```
5
6
7
8
```

6406532468349. ✖ None of these

✔ Official Answer Key: Option (A)

SC_067 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 250 Question Id : 640653737532 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

```
$ a=test
$ b=10
$ c=$((a + b))
$ d=$((a + c))
$ echo $d
```

What will be the value of `d` at the end of execution?

Options :

6406532468350. ✖ 10

6406532468351. ✔ 20

6406532468352. ✖ test1010

6406532468353. ✖ will result in error

✔ Official Answer Key: Option (B)

SC_069 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 252 Question Id : 640653737536 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

How many soft links are present in the list?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

✔ **Official Answer Key (Numerical Value): 2**

SC_079 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 247 Question Id : 6406531113965 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Assume that we are in the home directory of `user1`.

Which of the following commands will give us the count of files, other than text files (with a `.txt` extension) present in the current directory?

Options :

6406533774714. ✔ `find /home/user1 -not -name *.txt | wc -l`

6406533774715. ✖ `find /home/user1 -name *.txt | wc -l`

6406533774716. ✖ `ls /home/user1 | grep -v "(txt)$"`

6406533774717. ✖ `ls /home/user1 | grep "(txt)$" | wc -l`

Sub-Section Number :	4
Sub-Section Id :	640653168578
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

SC_081 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 249 Question Id : 6406531113968 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following ways can be used run the `sleep` command in the background for 5 minutes?

Options :

6406533774726. ✖ `sleep 5;`

6406533774727. ✔ `sleep 5m &;`

6406533774728. ✔ `coproc sleep 300s;`

6406533774729. ✖ `sleep 5m;`

Sub-Section Number :	5
Sub-Section Id :	640653168579
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B), Option (C)

SC_083 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 251 Question Id : 6406531113970 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What does `total 8` represent in the output of `ls -l`?

Options :

6406533774730. ✖ The total number of files in the directory

6406533774731. ✔ The total number of blocks used by the files and directories

6406533774732. ✖ The total size of the directory in bytes

6406533774733. ✖ The total number of links to the directory

✔ Official Answer Key: Option (B)

SC_084 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 252 Question Id : 6406531113971 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following commands can be used to allow group users to edit **`script.sh`**?

Options :

6406533774734. ✔ `chmod g+w script.sh`

6406533774735. ✖ `chmod g+x script.sh`

6406533774736. ✖ `chmod g-w script.sh`

6406533774737. ✖ `chmod 700 script.sh`

✔ Official Answer Key: Option (A)

SC_086 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 254 Question Id : 6406531113973 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The output of the command

`tail -n 10 example.txt` is

Options :

6406533774742. ✖ the first 10 characters of the file `example.txt`

6406533774743. ✖ the first 10 lines of the file `example.txt`

6406533774744. ✖ the last 10 characters of the file `example.text`

6406533774745. ✔ the last 10 lines of the file `example.txt`

✔ **Official Answer Key: Option (D)**

SC_088 | [MCQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 236 Question Id : 640653577993 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

```
mkdir mydir
touch mydir/file1.txt;touch mydir/file2.txt
mkdir mydir/subdir;touch mydir/subdir/file3.txt
rmdir mydir/subdir
```

Select the output of the above script.

Options :

6406531930127. ✖ remove directory `subdir`

6406531930128. ✖ remove directory `dir` and `subdir`

6406531930129. ✖ remove files from `subdir`

6406531930130. ✔ exit with exit code (not 0)

Sub-Section Number :	3
Sub-Section Id :	64065382639
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (D)

SC_095 | [MCQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 243 Question Id : 640653577999 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7

Question Label : Multiple Choice Question

Which of the following command will generate output for both `test.out` and `test.err` files. Assume that the `ech` is not a command and will generate a error message.

Options :

6406531930151. ✖ `(echo "test" && ech) > test.err 2>&1 | tee -a test.out`

6406531930152. ✔ `(echo "test" && ech) 2> test.err | tee >test.out`

6406531930153. ✖ `(echo "test" && ech) > test.err 2>&1 | tee >test.out`

6406531930154. ✖ `$(echo "test" && ech) > test.err 2>&1 | tee >test.out`

Sub-Section Number :	5
Sub-Section Id :	64065382641
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (B)

SC_101 | [SA] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 249 Question Id : 640653578003 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7

Question Label : Short Answer Question

How many background processes are running after the end of execution of the following script?

```
for i in {1..11}; do
        sleep 10 &
done

for i in {1..11..2}; do
        kill % # kill the last background process created
done
```

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number :	7
Sub-Section Id :	64065382643
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key (Numerical Value): 5

SC_102 | [SA] | Marks: +3.0 | *Quiz1 May23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 250 Question Id : 640653578006 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Short Answer Question

What will be the output of the following script?

```
read -n 4 line < <(echo abcdef)
echo $line$line
```

Hint:

```
$ cat < <(echo 123456) # send the stdout from echo command
to cat as stdin
123456

$ help read
read: read [-ers] [-a array] [-d delim] [-i text] [-n
nchars] [-N nchars] [-p prompt] [-t timeout] [-u fd] [name
...]
    Read a line from the standard input and split it into
fields.

    Reads a single line from the standard input, or from
file descriptor FD
    if the -u option is supplied.  The line is split into
fields as with word
    splitting, and the first word is assigned to the first
NAME, the second
    word to the second NAME, and so on, with any leftover
words assigned to
    the last NAME.  Only the characters found in $IFS are
recognized as word
    delimiters.
```

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

SC_105 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 161 Question Id : 640653825169 Question Type : MCQ

Correct Marks : 6

Question Label : Multiple Choice Question

What will be the output of the following command?

```
echo {A,B{1,2},C}
```

Options :

6406532775167. ✔ A B1 B2 C

6406532775168. ✖ A B 1 2 C

6406532775169. ✖ A {B1 B2} C

6406532775170. ✖ {A B1 B2 C}

✔ Official Answer Key: Option (A)

SC_106 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+5 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 162 Question Id : 640653825170 Question Type : MCQ
Correct Marks : 6
Question Label : Multiple Choice Question

Which of the following actions will delete the 7th line and **save and exit** in a `vi` editor or `emacs` editor?

Options :

```
$ vi filename.txt
:7 # all following commands in vi editor
dd #delete the line
:wq
```

```
$ emacs filename.txt
M-g g 7 # go to the 7th line
C-k # delete the line
C-x C-s # save the file
C-x C-c # exit Emacs
```

6406532775171. ✔

6406532775172. ✖

```
$ vi filename.txt
7 # all following commands in vi editor
D #delete the line
:wq
```

```
$ emacs filename.txt
C-x 7 # go to the 7th line
C-k # delete the line
C-x C-s # save the file
C-x C-c # exit Emacs
```

```
$ vi filename.txt
:7 # all following commands in vi editor
D #delete the line
:wq!
```

```
$ emacs filename.txt
M-7 # go to the 7th line
C-d # delete the line
C-x C-s # save the file
C-x C-c # exit Emacs
```

6406532775173. ✖

```
$ vi filename.txt
:7 # all following commands in vi editor
dd #delete the line
:w
```

```
$ emacs filename.txt
M-g g 7 # go to the 7th line
C-d # delete the line
C-x C-s # save the file
C-x C-c # exit Emacs
```

6406532775174. ✖

Sub-Section Number :	3
Sub-Section Id :	640653122772
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

SC_117 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 173 Question Id : 640653825165 Question Type : SA
Correct Marks : 8

Question Label : Short Answer Question

Given the following contents of a file `websites.txt`, what will be the output of the command `cat websites.txt | cut -d/ -f3 | grep -v 'www\.'|tail -1` ?

Hint

- For command `grep`
  -v, --invert-match
  Invert the sense of matching, to select non-matching lines.
- For command `cut`
  -d, --delimiter=DELIM
  use DELIM instead of TAB for field delimiter
  -f, --fields=LIST
  select only these fields; also print any line that contains no delimiter character, unless the -s option is specified

```
https://www.example.com/page1
http://www.test.org/page2
https://sub.domain.net/page3
http://anotherexample.com/page4
https://www.site.co.uk/page5
http://example.org/page6
```

Response Type : Alphanumeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Answers Case Sensitive : No
Text Areas : PlainText
Possible Answers :
example.org

CT

✔ **Official Answer Key (Numerical Value): example.org**

SC_120 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 132 Question Id : 6406531527866 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You want to count files in `/home/user/Data` that start with `sales_` followed by a single lowercase letter and end with `.csv`, e.g., `sales_a.csv`.

Which command will correctly give the count?

Options :

6406535115735. ✖ `ls /home/user/Data/sales_*.csv | wc -l`

6406535115736. ✖ `ls /home/user/Data/sales_?.csv | wc -l`

6406535115737. ✖ `ls /home/user/Data/sales_[a-z]?.csv | wc -l`

6406535115738. ✔ `ls /home/user/Data/sales_[a-z].csv | wc -l`

✔ Official Answer Key: Option (D)

SC_121 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+8 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 133 Question Id : 6406531527867 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which command will give information on open network ports on the machine?

Options :

6406535115739. ✔ `netstat`

6406535115740. ✖ `whois`

6406535115741. ✖ `wireshark`

6406535115742. ✖ `nslookup`

✔ Official Answer Key: Option (A)

SC_129 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+8 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 141 Question Id : 6406531527875 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which command is used to **display the IP address and network interfaces** of your Ubuntu system?

Options :

6406535115771. ✖ `ping`

6406535115772. ✔ `ifconfig`

6406535115773. ✖ `netstat`

6406535115774. ✖ `ipconf`

6406535115775. ✔ `ip addr`

Sub-Section Number :	5
Sub-Section Id :	640653244933
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B), Option (E)

SC_135 | [MCQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025 (+5 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 122 Question Id : 6406531525135 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You want to count files in `/home/user/Data` that start with `sales_` followed by a single lowercase letter and end with `.csv`, e.g., `sales_a.csv`.

Which command will correctly give the count?

Options :

6406535108098. ✖ `ls /home/user/Data/sales_*.csv | wc -l`

6406535108099. ✖ `ls /home/user/Data/sales_?.csv | wc -l`

6406535108100. ✖ `ls /home/user/Data/sales_[a-z]?.csv | wc -l`

6406535108101. ✔ `ls /home/user/Data/sales_[a-z].csv | wc -l`

✔ **Official Answer Key: Option (D)**

SC_150 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 230 Question Id : 640653668677 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following options contains command(s) which will output calendar for October and November of 2023.

Options :

6406532240235. ✖ `echo cal\ {10,11}\ 2023\;`

6406532240236. ✖ `$(echo cal\ {10,11}\ 2023\;)`

6406532240237. ✔ `echo cal\ {10,11}\ 2023\;|bash`

6406532240238. ✖ `$($(echo cal\ {10,11}\ 2023\;))`

6406532240239. ✔ `eval "$(echo cal\ {10,11}\ 2023\;)"`

Sub-Section Number :	6
Sub-Section Id :	64065395217
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (C), Option (E)

SC_152 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 232 Question Id : 640653668673 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Short Answer Question

At the end of the execution of the following script what will be the last value of `val1`.

```
$ cat .myscript.sh
#!/bin/bash

echo "Using \$*:"
for arg in "$*"; do
  val1=$arg
  echo $val1
done

echo "Using \$@:"
for arg in $@; do
  val2=$arg
  echo $val2
done

$ ./myscript.sh 11 22 33
```

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

11 22 33

✔ Official Answer Key (Numerical Value): 11 22 33

SC_154 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 234 Question Id : 640653668676 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Short Answer Question

What is the output of the following command?

```
$ echo $(( 10 * 10 - 5 * 4 ))
```

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

80

Sub-Section Number :	8
Sub-Section Id :	64065395219
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (Numerical Value): 80**

SC_159 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 135 Question Id : 640653995575 Question Type : MCQ

Correct Marks : 6

Question Label : Multiple Choice Question

What will be the output of the following command?

```
echo {1..3}-file_{A..C}
```

Options :

6406533361046. ✔ `1-file_A 1-file_B 1-file_C 2-file_A 2-file_B 2-file_C 3-file_A 3-file_B 3-file_C`

6406533361047. ✖ `1 2 3 - file A B C`

6406533361048. ✖ `{1..3}-file_{A..C}`

6406533361049. ✖ `1..3-file_A..C`

✔ **Official Answer Key: Option (A)**

SC_163 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 140 Question Id : 640653995569 Question Type : MSQ

Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following command sequences will run two commands in the background and print their PIDs?

Options :

6406533361025. ✔ `command1 & PID1=$!; command2 & PID2=$!; echo $PID1 $PID2`

6406533361026. ✖ `command1 & command2 & echo $!`

6406533361027. ✔ `command1 & PID1=$! && command2 & PID2=$! && echo $PID1 $PID2`

6406533361028. ✖ `command1 &; PID1=$!; command2 &; PID2=$!; echo $PID1 $PID2`

✔ **Official Answer Key: Option (A), Option (C)**

SC_168 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 145 Question Id : 6406539955778 Question Type : MSQ
Correct Marks : 6 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which command will create a directory `data` with a subdirectory `backup` and set the permissions so that the owner has full control, the group can read and execute, and others have no permissions, then verify the permissions?

Hint



Options :

```
mkdir -p data/backup
chmod 750 data data/backup
ls -ld data data/backup
```

6406533361058. ✔

6406533361059. ✔

✔ Official Answer Key: Option (A), Option (B)

SC_171 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 148 Question Id : 640653995571 Question Type : SA

Correct Marks : 8

Question Label : Short Answer Question

Given the following contents of a file `websites.txt`, what will be the output of the command

```
cat websites.txt | cut -d: -f1 | sort | uniq| tail -1 ?
```

```
https://www.example.com/page1 http://www.test.org/page2
https://sub.domain.net/page3 http://anotherexample.com/page4
https://www.site.co.uk/page5 http://example.org/page6
```

NOTE: Enter the exact answer without any extra space in the beginning or at the end.

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : Yes

Text Areas : PlainText

Possible Answers :

https

AppDev2

✔ **Official Answer Key (Numerical Value): https**

SC_172 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 145 Question Id : 640653995860 Question Type : MSQ
Correct Marks : 6 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which command will create a directory `data` with a subdirectory `backup` and set the permissions so that the owner has full control, the group can read and execute, and others have no permissions, then verify the permissions?

Hint



Options :

```
mkdir -p data/backup
chmod 750 data data/backup
ls -ld data data/backup
```

6406533361842. ✔

6406533361843. ✔

✔ Official Answer Key: Option (A), Option (B)

SC_173 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 145 Question Id : 6406539961 42 Question Type : MSQ
Correct Marks : 6 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which command will create a directory `data` with a subdirectory `backup` and set the permissions so that the owner has full control, the group can read and execute, and others have no permissions, then verify the permissions?

Hint



Options :

```
mkdir -p data/backup
chmod 750 data data/backup
ls -ld data data/backup
```

6406533362626. ✔

6406533362627. ✔

✔ Official Answer Key: Option (A), Option (B)

SC_180 | [MCQ] | Marks: +3.0 | *System commands*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 8 Question Id : 6406532159056 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Match the following Vi/Vim and Emacs commands to their correct functionality.

Vi / Vim Commands

Command	Use
1. `dd`	**A.** Search forward for a specific pattern
2. `u`	**B.** Paste the text present in the buffer
3. `p`	**C.** Delete the current line
4. `/`	**D.** Undo the last structural change

Emacs Commands

Note: Here, `C-` means the `Ctrl` key and `M-` means the `Alt` (Meta) key.

Command	Use
1. `C-x C-s`	**A.** Cut (kill) the selected text region
2. `C-s`	**B.** Save the current buffer/file
3. `C-w`	**C.** Paste (yank) previously cut text
4. `C-y`	**D.** Search forward incrementally

Which of the following option combinations is correct?
Options :
6406537023172. ✔ **Vi:** 1-C, 2-D, 3-B, 4-A | **Emacs:** 1-B, 2-D, 3-A, 4-C
6406537023173. ✖ **Vi:** 1-C, 2-D, 3-A, 4-B | **Emacs:** 1-B, 2-A, 3-D, 4-C
6406537023174. ✖ **Vi:** 1-D, 2-C, 3-B, 4-A | **Emacs:** 1-C, 2-D, 3-A, 4-B
6406537023175. ✖ **Vi:** 1-D, 2-C, 3-A, 4-B | **Emacs:** 1-C, 2-A, 3-D, 4-B

✔ Official Answer Key: Option (A)

SC_184 | [MCQ] | Marks: +3.0 | *System commands*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 12 Question Id : 6406532159063 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

```
$ nslookup google.com
```

The above command is used to

Options :

6406537023200. ✔ retrieve information about a domain name, such as its server and IP address.

6406537023201. ✖ looks up the number of pings to the domain name "google.com".

6406537023202. ✖ Gets the DNS of the domain on google.com

6406537023203. ✖ none of these

✔ Official Answer Key: Option (A)

SC_185 | [MCQ] | Marks: +3.0 | *System commands*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.1 OS Architecture, Linux Kernel & Shell Basics

Question Number : 13 Question Id : 6406532159064 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
An operator attempts to run a critical update sequence on a production server using the following two independent commands:

- **Command 1:** `./verify_env.sh && ./run_update.sh`
- **Command 2:** `./check_disk.sh || ./clean_tmp.sh`

After running both commands, the engineer notices that `./run_update.sh` **did not execute**, but `./clean_tmp.sh` **did execute**.

Based on the short-circuit behavior of shell operators, what can be definitively concluded about the exit statuses of the initial scripts (`./verify_env.sh` and `./check_disk.sh`)?
Options :
6406537023204. ✖ `./verify_env.sh` exited with a success status (0), and `./check_disk.sh` exited with a failure status (non-zero).

6406537023205. ✖ `./verify_env.sh` exited with a failure status (non-zero), and `./check_disk.sh` exited with a success status (0).

6406537023206. ✖ Both `./verify_env.sh` and `./check_disk.sh` exited with a success status (0).

6406537023207. ✔ Both `./verify_env.sh` and `./check_disk.sh` exited with a failure status (non-zero).

Sub-Section Number :	5
Sub-Section Id :	640653396307
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (D)

• 1.4 Viewing File Contents (2 Questions)

SC_007 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.4 Viewing File Contents

Question Number : 260 Question Id : 640653345411 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Select Question

```
$ cat file1.txt
1
2
3
4
5
6
7
8
```

Which of the following commands will give the output as.

```
6
7
```

Options :

6406531148814. ✔ `cat file1.txt | head -7 | tail -2`

6406531148815. ✖ `cat file1.txt | head -6 | tail -7`

6406531148816. ✖ `cat file1.txt | tail -6 | head -3`

6406531148817. ✔ `cat file1.txt | tail -3 | head -2`

✔ Official Answer Key: Option (A), Option (D)

SC_015 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.4 Viewing File Contents

Question Number : 268 Question Id : 640653345421 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

What will the contents of the file `notes.txt` be after executing the below commands?

```
$ echo note > notes.txt
$ echo memo > memos.txt
$ echo list > lists.txt
$ cat lists.txt memos.txt >> notes.txt
```

Options :

6406531148850. ✖
```
note
```

6406531148851. ✖
```
memo
list
note
```

6406531148852. ✖
```
list
memo
note
```

6406531148853. ✔
```
note
list
memo
```

6406531148854. ✖
```
note
memo
list
```

✔ Official Answer Key: Option (D)

• 1.5 File Metadata & Listing Attributes (12 Questions)

SC_027 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 228 Question Id : 6406531298622 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

To display the first 7 lines and the last 3 lines of a file named `config.txt`, which combination of commands is correct?

Options :

6406534366687. ✔ `head -n 7 config.txt` and `tail -n 3 config.txt`

6406534366688. ✖ `cat -first 7 config.txt` and `cat -last 3 config.txt`

6406534366689. ✖ `less +7 config.txt` and `less -G3 config.txt`

6406534366690. ✖ `display -s 1-7 config.txt` and `display -e 1-3 config.txt`

✔ **Official Answer Key: Option (A)**

SC_048 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 249 Question Id : 640653470183 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7

Question Label : Multiple Choice Question

When will the script `/home/rohit/premodel.sh` get executed.

```
5 0 * 8 * /home/rohit/premodel.sh
```

Hint: Below is the description of the sequence in the cron job command. It tells at what date/time periodically the job needs to be executed.

```
*   *   *   *   *   <Command(s) with argument>
|   |   |   |   |             |
|   |   |   |   |      Command or Script to Execute
|   |   |   |   |
|   |   |   |   |
|   |   |   |   |
|   |   |   | Day of the Week(0-6)
|   |   |   |
|   |   | Month of the Year(1-12)
|   |   |
|   | Day of the Month(1-31)
|   |
| Hour(0-23)
|
Min(0-59)
```

Options :

6406531562462. ✔ Everyday at 00:05 in August

6406531562463. ✖ Everyday at 05:00 in August

6406531562464. ✖ Everyday at 08:00 in May

6406531562465. ✖ Everyday at 08:05 in May

✔ Official Answer Key: Option (A)

SC_053 | [MSQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 254 Question Id : 640653470188 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8 Selectable Option : 0

Question Label : Multiple Select Question

Choose the type of user(s) who can write to the file traceback.txt using the console output given below. [MSQ]

```
$ ls -l
-rw-rw-r--  1 abdul student     6977 Aug  8 19:14  traceback.txt
```

Options :

6406531562479. ✔ abdul

6406531562480. ✔ ankit who belong to the group "student"

6406531562481. ✖ catherine who do not belong to the group "student"

6406531562482. ✖ jina who belong to the group "abdul"

Sub-Section Number :	6
Sub-Section Id :	64065367733
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A), Option (B)

SC_071 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 254 Question Id : 640653737538 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

From the given information, which of the following types are present in the list?

Options :

6406532468365. ✖ regular file

6406532468366. ✔ directory

6406532468367. ✖ hard links

6406532468368. ✔ soft/symbolic links

6406532468369. ✔ character files

6406532468370. ✖ block files

6406532468371. ✖ named pipe files

6406532468372. ✖ socket files

Sub-Section Number :	7
Sub-Section Id :	640653107685
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (B), Option (D), Option (E)

SC_085 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*
Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 253 Question Id : 6406531113972 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Which of the following represents the correct file size of `example.txt` ?

Options :

6406533774738. ✖ 96 blocks

6406533774739. ✔ 96 bytes

6406533774740. ✖ 96 kilobytes

6406533774741. ✖ 96 bits

✔ Official Answer Key: Option (B)

SC_123 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+8 more papers)*
Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 135 Question Id : 6406531527869 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

You are troubleshooting a script `deploy.sh` and want to:

- Show normal output on the screen
- Save **only error messages** to a file called `errors.txt`

Which command will achieve this?

Options :

6406535115747. ✖ `./deploy.sh > errors.txt`

6406535115748. ✖ `./deploy.sh > /dev/null`

6406535115749. ✔ `./deploy.sh 2> errors.txt`

6406535115750. ✖ `./deploy.sh &> errors.txt`

✔ Official Answer Key: Option (C)

SC_132 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+8 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 144 Question Id : 6406531527879 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

You want to **create a new directory** named `Reference` inside `Resources`, and inside that, create an empty file `index.md`. Which of the following command sets will do this correctly?

Options :

6406535115778. ✔

```
mkdir Resources/Reference
touch Resources/Reference/index.md
```

6406535115779. ✔

```
mkdir -p Resources/Reference
cd Resources/Reference
touch index.md
```

6406535115780. ✖

```
mkdir Resources/index.md
touch Resources/Reference
```

6406535115781. ✖

```
mkdir Reference
touch Reference/index.md
```

✔ Official Answer Key: Option (A), Option (B)

SC_133 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+8 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 145 Question Id : 6406531527880 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

You want to perform the following operations on the structure given in common statement:

1. **Move** the file `main.py` from `ProjectA` to `ProjectB`
2. **Delete** the now-empty `ProjectA` directory
3. **List** the contents of `ProjectB` to confirm the move

Which of the following command sets **correctly** achieves all three steps?

Options :

6406535115782. ✔

```
mv Projects/ProjectA/main.py Projects/ProjectB/
rmdir Projects/ProjectA
ls Projects/ProjectB
```

6406535115783. ✖

```
mv Projects/ProjectA/main.py Projects/ProjectB/
rm Projects/ProjectA
ls Projects/ProjectB
```

6406535115784. ✖

```
cp Projects/ProjectA/main.py Projects/ProjectB/
rmdir Projects/ProjectA
ls Projects/ProjectA
```

6406535115785. ✔

```
mv Projects/ProjectA/main.py Projects/ProjectB/
rm -rf Projects/ProjectA
ls Projects/ProjectB
```

✔ Official Answer Key: Option (A), Option (D)

SC_137 | [MCQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025 (+5 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 126 Question Id : 6406531525139 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You want to:

- Combine the contents of two files: `intro.txt` and `summary.txt`
- Sort all lines alphabetically
- Save the final sorted output to a file named `full_report.txt`, **overwriting** any existing content

Which of the following commands correctly performs **all** of these steps?

Options :

6406535108114. ✖ `cat intro.txt | cat summary.txt | sort > full_report.txt`

6406535108115. ✖ `cat intro.txt summary.txt | sort >> full_report.txt`

6406535108116. ✖ `cat intro.txt > full_report.txt && cat summary.txt | sort`

6406535108117. ✔ `cat intro.txt summary.txt | sort > full_report.txt`

✔ Official Answer Key: Option (D)

SC_179 | [MCQ] | Marks: +3.0 | *System commands*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 7 Question Id : 6406532159055 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A DevOps engineer is configuring secure, passwordless SSH access from a local administration workstation to a remote Linux cloud server. Which of the following operations is correctly described regarding the placement and management of the keys?

Options :

6406537023168. ✖ The private key must be appended to the `~/.ssh/authorized_keys` file on the remote server so it can decrypt incoming client challenges.

6406537023169. ✖ Both the public and private keys must be copied to the remote server's `/etc/ssh/ssh_host_key` directory to establish trusted mutual identity.

6406537023170. ✔ The public key must be securely transferred and appended to the `~/.ssh/authorized_keys` file on the remote server, while the private key remains exclusively on the local workstation.

6406537023171. ✖ The private key is automatically deleted from the local workstation by the `ssh-copy-id` utility once the matching public key is uploaded to the destination server.

✔ Official Answer Key: Option (C)

SC_187 | [MCQ] | Marks: +3.0 | *System commands*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 15 Question Id : 6406532159065 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question
Consider the following standard IPv4 address class ranges:

- Class A: `1.0.0.0 – 126.255.255.255`
- Class B: `128.0.0.0 – 191.255.255.255`
- Class C: `192.0.0.0 – 223.255.255.255`
- Loopback: `127.0.0.0 – 127.255.255.255`

A file named `ips.txt` contains a list of various IP addresses. You want to extract **only the lines containing Class B IP addresses** based strictly on their first octet value (ranging from 128 to 191).

Which of the following commands will correctly filter and print these lines?
Options :

6406537023208. ✔ `grep -E '^(12[8-9]|1[3-8][0-9]|19[0-1])\.' ips.txt`

6406537023209. ✖ `grep -E '^(128|191)\.' ips.txt`

6406537023210. ✖ `grep -E '^([128-191])\.' ips.txt`

6406537023211. ✖ `grep -E '^(12[8-9]|[1-3][0-8][0-9]|19[0-1])\.' ips.txt`

Sub-Section Number :	7
Sub-Section Id :	640653396309
Question Shuffling Allowed :	No

Question Id : 6406532159066 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (16 to 17)

✔ Official Answer Key: Option (A)

SC_189 | [MCQ] | Marks: +3.0 | *System commands*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.5 File Metadata & Listing Attributes

Question Number : 17 Question Id : 6406532159068 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
An engineer executes the following command sequence in their terminal, starting from `/home/developer`:

```
cp test/test_core.py build/
rmdir test
ls build
```

Which of the following statements correctly describes the final outcome and system state after these commands are run?
Options :

6406537023216. ✗ The `test_core.py` file is moved to `build/`, the empty `test` directory is successfully deleted, and `ls build` lists `test_core.py`.

6406537023217. ✗ The command sequence throws a syntax error on the first line because `build/` must be specified as an absolute path.

6406537023218. ✔ A copy of `test_core.py` is created in `build/`, but the `rmdir` command fails with an error because the `test` directory still contains the original file.

6406537023219. ✗ The entire `test` directory and its contents are forcefully removed, leaving `build/` completely empty.

✔ Official Answer Key: Option (C)

• 1.3 File & Directory Operations (6 Questions)

SC_043 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.3 File & Directory Operations

Question Number : 244 Question Id : 640653470182 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

Select the missing command.

```
$ touch a
$ echo 5 > a
$ ** MISSING COMMAND **
$ cat a b
5
apples
```

Options :

6406531562458. ✖ touch b

6406531562459. ✖ echo b

6406531562460. ✔ echo apples > b

6406531562461. ✖ echo apples

✔ **Official Answer Key: Option (C)**

SC_047 | [SA] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.3 File & Directory Operations

Question Number : 248 Question Id : 640653470187 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Short Answer Question

How many files will be created at the end of execution by the below script?

```
touch a b c d
touch a e d
echo $RANDOM >> b
```

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number :	4
Sub-Section Id :	64065367731
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (Numerical Value): 5**

SC_073 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.3 File & Directory Operations

Question Number : 241 Question Id : 6406531113959 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Assume that the user is in the home directory `/home/user1`. Consider the following commands.

```
mkdir app; mkdir app/public
cd app || cd app/public
touch index.html
cd ..
mv image.png app/public
```

What will be the current working directory after execution of the above commands?

Options :

6406533774690. ✖ `/home/user1/app/public`

6406533774691. ✔ `/home/user1`

6406533774692. ✖ `/home/user1/app`

6406533774693. ✖ `/app`

✔ Official Answer Key: Option (B)

SC_103 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.3 File & Directory Operations

Question Number : 159 Question Id : 640653825160 Question Type : MCQ

Correct Marks : 6

Question Label : Multiple Choice Question

```
mkdir mydir
touch mydir/file1.txt; touch mydir/file2.txt
mkdir mydir/subdir mydir/subdir1;  touch mydir/subdir1/file3.txt
rmdir mydir/subdir
```

Select the output from the above script.

Options :

6406532775134. ✖ remove directory subdir

6406532775135. ✖ remove directory `dir` and `subdir`

6406532775136. ✔ remove directory `subdir`

6406532775137. ✖ exit with error code

✔ Official Answer Key: Option (C)

SC_130 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+8 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.3 File & Directory Operations

Question Number : 142 Question Id : 6406531527876 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider the command:

```
1 mkdir test-{0..3} && mkdir test-{4..6} || mkdir test-{7..10}
```

How many directories will be created if none of the directories exist initially?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

7

Sub-Section Number :	6
Sub-Section Id :	640653244934
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 7**

SC_158 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 1: Introduction to Operating Systems, Shell & Basic Commands › 1.3 File & Directory Operations

Question Number : 134 Question Id : 640653995566 Question Type : MCQ
Correct Marks : 6
Question Label : Multiple Choice Question

```
mkdir -p site/public; touch site/public/index.html
touch site/public/style.css
mkdir site/private site/static; touch site/static/image.png
rmdir site/static
```

Select the output from the above script.

Hint

```
```bash
MKDIR(1)                                                    User
Commands                                                    MKDIR(1)

NAME
       mkdir - make directories

SYNOPSIS
       mkdir [OPTION]... DIRECTORY...

DESCRIPTION
       Create the DIRECTORY(ies), if they do not already exist.

...
       -p, --parents
              no error if existing, make parent directories as needed
...
```

Options :

6406533361013. ✖ Remove directory `static`

6406533361014. ✖ Remove directory `site` and `static`

6406533361015. ✖ Remove files from `static`

6406533361016. ✔ Exit with error code

✔ Official Answer Key: Option (D)

Week 2: File System Hierarchy, Permissions, Redirection & Links (21 Questions)

• 2.1 File System Hierarchy Standard (FHS) (2 Questions)
```

SC_017 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.1 File System Hierarchy Standard (FHS)

Question Number : 270 Question Id : 640653345423 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

```
#!/usr/bin/bash

sleep 10 &
sleep 10 &
sleep 10 &
echo Good Morning
```

After how many seconds the text `Good Morning` will be displayed?

Options :

6406531148860. ✖ After 10 seconds.

6406531148861. ✖ After 20 seconds.

6406531148862. ✖ After 30 seconds.

6406531148863. ✔ Immediately.

✔ Official Answer Key: Option (D)

SC_038 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.1 File System Hierarchy Standard (FHS)

Question Number : 239 Question Id : 6406531298633 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements accurately describe parts of the `ls -lai /usr/bin` command and its expected output? (Select all that apply)

Options :

6406534366731. ✔ `ls` is the command to list directory contents.

6406534366732. ✖ `-l` option enables a short listing format.

6406534366733. ✖ `-a` option ensures that hidden files (starting with `.`) are not displayed.

6406534366734. ✔ `-i` option displays the inode number of each file.

6406534366735. ✔ The output would primarily consist of user command executable files.

Sub-Section Number :	4
Sub-Section Id :	640653206812
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (D), Option (E)

• 2.4 I/O Redirection & Standard Streams (7 Questions)

SC_018 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.4 I/O Redirection & Standard Streams

Question Number : 271 Question Id : 640653345428 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

```
$ var="IIT Madras"
```

The above command is executed before running any of the commands below. Match the commands below to their outputs.

Commands	Output
1. echo ${var:-"value"}	a. value
2. echo ${var:+"value"}	b. IIT Madras
3. echo ${var:="value"}	c. T Ma
4. echo ${var:2:4}	d. stem

Options :

6406531148884. ✖ 1—>b; 2—>a; 3—>a; 4—>c;

6406531148885. ✖ 1—>a; 2—>b; 3—>b; 4—>d;

6406531148886. ✖ 1—>a; 2—>b; 3—>b; 4—>c;

6406531148887. ✔ 1—>b; 2—>a; 3—>b; 4—>c;

Sub-Section Number : 7

Sub-Section Id : 64065348830

Question Shuffling Allowed : Yes

✔ Official Answer Key: Option (D)

SC_033 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.4 I/O Redirection & Standard Streams

Question Number : 234 Question Id : 6406531298628 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A command `generate_raw_data` produces a stream of data on its standard output (stdout) and occasionally prints error or status messages to its standard error (stderr). You need to pipe the raw data (stdout) to another command `process_stream` for further processing. Simultaneously, you want to capture only the error messages (stderr) from `generate_raw_data` into a file named `raw_data_errors.log`, overwriting the log file each time. The final processed data from `process_stream` should appear on the terminal.

Which command sequence correctly achieves this?

Options :

6406534366711. ✔ `generate_raw_data 2> raw_data_errors.log | process_stream`

6406534366712. ✖ `generate_raw_data | process_stream 2> raw_data_errors.log`

6406534366713. ✖ `generate_raw_data > process_stream 2> raw_data_errors.log`

6406534366714. ✖ `(generate_raw_data |& process_stream) 2> raw_data_errors.log`

✔ Official Answer Key: Option (A)

SC_037 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.4 I/O Redirection & Standard Streams

Question Number : 238 Question Id : 6406531298632 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

What are the potential consequences of using the `>` redirection operator (e.g., `command > file1`)?

Options :

6406534366727. ✖ The contents of `file1` will be appended to if it exists.

6406534366728. ✔ The contents of `file1` will be overwritten if it exists.

6406534366729. ✔ A new `file1` will be created if it does not exist.

6406534366730. ✖ The command will always ask for confirmation before modifying `file1`.

✔ Official Answer Key: Option (B), Option (C)

SC_078 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.4 I/O Redirection & Standard Streams

Question Number : 246 Question Id : 6406531113961 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following commands.

```
echo Hello > hello.txt
cat hello.txt /blah.txt > output.txt 2>> output.txt
```

After execution, which of the following represents the contents of `output.txt` ?

Options :

6406533774698. ✖

```
cat: /blah.txt: No such file or directory
```

6406533774699. ✔

```
Hello
cat: /blah.txt: No such file or directory
```

6406533774700. ✖

```
Hello
```

6406533774701. ✖

```
cat: /blah.txt: No such file or directory
Hello
```

✔ Official Answer Key: Option (B)

SC_090 | [MCQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.4 I/O Redirection & Standard Streams

Question Number : 238 Question Id : 640653577995 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

Identify the correct pair for the output of each command.

```
$ date
Mon Jun 12 12:28:02 IST 2023
$ date_disp=($(date)) # creates an array. each space
separated fields becomes an element.
$ echo $date_disp
$ echo $date_disp[5]
$ echo ${date_disp[5]}
$ echo ${date_disp:1:8}the
```

Sr.No	command	option	output
1	`echo $date_disp`	a	`onthe`
2	`echo ${date_disp[5]}`	b	`mon[5]`
3	`echo $date_disp[5]`	c	`2023`
4	`echo ${date_disp:1:8}the`	d	`Mon`

Options :

6406531930135. ✖ 1->b,2->a,3->c,4->d

6406531930136. ✖ 1->d, 2->b, 3 ->c, 4->a

6406531930137. ✔ 1->d, 2->c, 3->b, 4->a

6406531930138. ✖ none of these

✔ Official Answer Key: Option (C)

SC_114 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.4 I/O Redirection & Standard Streams

Question Number : 170 Question Id : 640653825162 Question Type : MSQ

Correct Marks : 7 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following commands will ensure that **both stdout and stderr** of a command are written to the same file?

Options :

6406532775142. ✔ `command > output.txt 2>&1`

6406532775143. ✖ `command 2>&1 > output.txt`

6406532775144. ✔ `command &> output.txt`

6406532775145. ✖ `command 1>&2 > output.txt`

Sub-Section Number :	6
Sub-Section Id :	640653122775
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

SC_160 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.4 I/O Redirection & Standard Streams

Question Number : 137 Question Id : 640653995568 Question Type : MCQ

Correct Marks : 8

Question Label : Multiple Choice Question

Which of the following command sequences will ensure that both stdout and stderr are written to `output.log` and also displayed in the terminal?

Options :

6406533361021. ✔ `command 2>&1 | tee output.log`

6406533361022. ✖ `command | tee output.log 2>&1`

6406533361023. ✖ `command &> output.log | tee output.log`

6406533361024. ✖ `command > output.log 2>&1 | tee output.log`

✔ **Official Answer Key: Option (A)**

- **2.6 Hard Links & Symbolic Links** (7 Questions)

SC_023 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.6 Hard Links & Symbolic Links

Question Number : 276 Question Id : 640653345427 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Select Question

Which of the following statements for hard and soft(symbolic) links are true.

Options :

6406531148880. ✔ Inode of a file and its hard link will be the same.

6406531148881. ✖ Inode of a file and its soft link will be the same.

6406531148882. ✖ Hard links become inaccessible after the file it links to is deleted.

6406531148883. ✔ Soft links become inaccessible after the file it links to is deleted.

Sub-Section Number :	9
Sub-Section Id :	64065348832
Question Shuffling Allowed :	No

Question Id : 640653345417 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (277 to 278)

✔ Official Answer Key: Option (A), Option (D)

SC_045 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.6 Hard Links & Symbolic Links

Question Number : 246 Question Id : 640653470190 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

A college student uses a pen drive to store files. The pen drive is always mounted on his system at `/mount` as a separate file system. His pen drive contains lots of files and directories. He wants to create a link to the directory `/mount/bio/phase-2/project-312` inside his home directory. What link can the student use?

Options :

6406531562487. ✔ soft link

6406531562488. ✖ hard link

6406531562489. ✖ either soft link or hard link

6406531562490. ✖ links will not work for this requirement

Sub-Section Number :	3
Sub-Section Id :	64065367730
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (A)**

SC_075 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.6 Hard Links & Symbolic Links

Question Number : 243 Question Id : 6406531113962 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider that we have a file named `example.txt` in the current working directory.

```
ln example.txt example_a.txt
ln -s example.txt example_b.txt
```

After execution of the commands above, a hard link and a soft link to `example.txt` is created as `example_a.txt` and `example_b.txt`, respectively.

Choose the appropriate option from the following.

Options :

6406533774702. ✖ The inode numbers of `example.txt` and `example_a.txt` are different.

6406533774703. ✖ The inode numbers of `example_a.txt` and `example_b.txt` are same.

6406533774704. ✔ The inode numbers of `example.txt` and `example_a.txt` are same.

6406533774705. ✖

None of these

✔ Official Answer Key: Option (C)

SC_109 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.6 Hard Links & Symbolic Links

Question Number : 165 Question Id : 640653825171 Question Type : MSQ

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which command will display the inode number of the file `myfile`?

Options :

6406532775175. ✔ `ls -i myfile`

6406532775176. ✖ `ls -l myfile`

6406532775177. ✔ `stat myfile`

6406532775178. ✖ `file myfile`

✔ Official Answer Key: Option (A), Option (C)

SC_118 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+5 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.6 Hard Links & Symbolic Links

Question Number : 130 Question Id : 6406531527864 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Suppose you run the following commands:

```
touch data.txt
ln data.txt hardlink_data
ln -s data.txt softlink_data
```

Which statement correctly describes the inode behavior of the links?

Options :

6406535115727. ✔ `hardlink_data` and `data.txt` share the same inode; `softlink_data` has a different inode

6406535115728. ✖ All three files have the same inode

6406535115729. ✖ `softlink_data` and `data.txt` share the same inode; `hardlink_data` has a different one

6406535115730. ✖ Each of the three files has a different inode

✔ Official Answer Key: Option (A)

SC_134 | [MCQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.6 Hard Links & Symbolic Links

Question Number : 120 Question Id : 6406531525133 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Suppose you run the following commands:

```
touch data.txt
ln data.txt hardlink_data
ln -s data.txt softlink_data
```

Which statement correctly describes the inode behavior of the links?

Options :

6406535108090. ✔ `hardlink_data` and `data.txt` share the same inode; `softlink_data` has a different inode

6406535108091. ✖ All three files have the same inode

6406535108092. ✖ `softlink_data` and `data.txt` share the same inode; `hardlink_data` has a different one

6406535108093. ✖ Each of the three files has a different inode

✔ **Official Answer Key: Option (A)**

SC_167 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.6 Hard Links & Symbolic Links

Question Number : 144 Question Id : 640653995577 Question Type : MSQ

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which command will display the inode number of the file `myfile` ?

Options :

6406533361054. ✔ `ls -i myfile`

6406533361055. ✖ `ls -l myfile`

6406533361056. ✔ `stat myfile`

6406533361057. ✖ `file myfile`

✔ **Official Answer Key: Option (A), Option (C)**

• 2.2 File Permissions & Ownership (5 Questions)

SC_041 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.2 File Permissions & Ownership

Question Number : 242 Question Id : 6406531298637 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

To change the permissions of `report.txt` (currently `-rw-r--r--`) so that only the owner can read and write, the group can read, and others have no permissions, which command is correct, and what will the new permission string be?

Options :

6406534366741. ✖ Command: `chmod 750 /home/user/Documents/report.txt`, New string: `-rwxr-x---`

6406534366742. ✔ Command: `chmod 640 /home/user/Documents/report.txt`, New string: `-rw-r-----`

6406534366743. ✖ Command: `chmod 740 /home/user/Documents/report.txt`, New string: `-rw-r-----`

6406534366744. ✔ Command: `chmod u=rw,g=r,o= /home/user/Documents/report.txt`, New string: `-rw-r-----`

✔ Official Answer Key: Option (B), Option (D)

SC_059 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.2 File Permissions & Ownership

Question Number : 242 Question Id : 640653737529 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

For a log file 'file.txt' the user wants to change the permissions such that he gives read, write, and execute to the owner; read and execute to the group and others. which of the following commands will accomplish the same?

Options :

6406532468337. ✖ `chmod u+rwx,g+rw,o+rw file.txt`

6406532468338. ✔ `chmod u+rwx,g+rx,o+rx file.txt`

6406532468339. ✖ `chmod 644 file.txt`

6406532468340. ✔ `chmod 755 file.txt`

✔ Official Answer Key: Option (B), Option (D)

SC_087 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.2 File Permissions & Ownership

Question Number : 255 Question Id : 6406531113974 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Select the appropriate option(s) regarding permissions of `script.sh`

Options :

6406533774746. ✖ All users can execute `script.sh`

6406533774747. ✔ All users can read `script.sh`

6406533774748. ✔ Group users can execute `script.sh`

6406533774749. ✖ Group users can edit `script.sh`

✔ Official Answer Key: Option (B), Option (C)

SC_098 | [MSQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.2 File Permissions & Ownership

Question Number : 246 Question Id : 640653578001 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

Given the following input identify which of the statements present in options are TRUE.

```
$ ls -li

   23 -rwxr-xr-x 3 root root  35K Jul  4  2019 bunzip2
31611 -rwxr-xr-x 1 root root 2.0M Nov 25  2021 busybox
   23 -rwxr-xr-x 3 root root  35K Jul  4  2019 bzcat
   25 lrwxrwxrwx 1 root root    6 Jul  4  2019 bzcmp ->
bzdiff
   26 -rwxr-xr-x 1 root root 2.1K Jul  4  2019 bzdiff
   27 lrwxrwxrwx 1 root root    6 Jul  4  2019 bzegr ->
bzgr
   28 -rwxr-xr-x 1 root root 4.8K Jul  4  2019 bzexe
   29 lrwxrwxrwx 1 root root    6 Jul  4  2019 bzfgr ->
bzgr
   30 -rwxr-xr-x 1 root root 3.6K Jul  4  2019 bzgrep
   23 -rwxr-xr-x 3 root root  35K Jul  4  2019 bzip2
   31 -rwxr-xr-x 1 root root  14K Jul  4  2019
bzip2recover
   32 lrwxrwxrwx 1 root root    6 Jul  4  2019 bzl -> bzm
   33 -rwxr-xr-x 1 root root 1.3K Jul  4  2019 bzmore
```

Options :

6406531930156. ✖ files `bzegr` and `bzfgr` are hard links

6406531930157. ✔ files `bzegr` and `bzfgr` are soft links

6406531930158. ✖ files `bzl` and `bzm` are hard links

6406531930159. ✔ files `bzcat` and `bzip2` are hard links

✔ Official Answer Key: Option (B), Option (D)

SC_107 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 2: File System Hierarchy, Permissions, Redirection & Links › 2.2 File Permissions & Ownership

Question Number : 163 Question Id : 640653825166 Question Type : MSQ

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

You try to SSH into a remote VM using the command `ssh user@remote-vm` and receive the following error:

```
Permission denied (publickey).
```

Which of the following option(s) are likely the cause?

Options :

6406532775155. ✔ The SSH key is not added to the SSH agent.

6406532775156. ✖ The remote VM's SSH server is down.

6406532775157. ✔ The permissions on the SSH key file are too open.

6406532775158. ✔ The username is incorrect.

✔ **Official Answer Key: Option (A), Option (C), Option (D)**

Week 3: Text Processing Filters, Grep/Regex, Sed & Awk (53 Questions)

- **3.3 Pattern Searching with Grep & Regular Expressions** (30 Questions)

SC_010 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 263 Question Id : 640653345426 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

What does the output of the command `ls -lR ./ | grep -e "^[d]" | wc -l` represents?

Hint: Command `ls -lR` will recursively list all the files and subdirectories in the long listing format.

Options :

6406531148875. ✖ Number of files in the current directory.

6406531148876. ✖ Number of directories in the current directory.

6406531148877. ✖ Number of directories and all sub-directories in the parent directory.

6406531148878. ✔ Number of directories and all sub-directories in the current directory.

6406531148879. ✖ Number of directories in the parent directory.

Sub-Section Number :	4
Sub-Section Id :	64065348827
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (D)

SC_011 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 264 Question Id : 640653345414 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Below are three date-time formats according to ISO 8601 in UTC

```
2022-05-06T13:00:56+00:00
2022-05-06T13:00:56Z
20220506T130056Z
```

Select the extended regular expression that matches only with one of the above date-time formats.

Options :

6406531148827. ✖ `^[0-9]{4}-*[0-9]{2}-*[0-9]{2}T[0-9]{2}:*[0-9]{2}:*[0-9]{2}(Z|\+00:00)$`

6406531148828. ✖ `^[0-9]{4}-+[0-9]{2}-+[0-9]{2}T[0-9]{2}:+[0-9]{2}:+[0-9]{2}(Z|\+00:00)$`

6406531148829. ✔ `^[0-9]{4}-?[0-9]{2}-?[0-9]{2}T[0-9]{2}:?[0-9]{2}:?[0-9]{2}(Z|\+00:00)$`

6406531148830. ✖ `^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}(Z|\+00:00)$`

Sub-Section Number :	5
Sub-Section Id :	64065348828
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (C)

SC_034 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 235 Question Id : 6406531298629 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If you use the pattern `^A.*Z$` with `grep` (defaulting to BRE), what kind of lines would it match?

Options :

6406534366715. ✖ Lines that contain "A" followed by any character, then "Z" anywhere on the line.

6406534366716. ✔ Lines that start with "A", have zero or more of any character (except newline) in between, and end with "Z".

6406534366717. ✖ Lines that contain either "A" or "Z" at the very beginning or very end of the line.

6406534366718. ✖ Lines that consist only of the characters "A" and "Z" with nothing in between.

✔ Official Answer Key: Option (B)

SC_035 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 236 Question Id : 6406531298630 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a file named `data.txt` with the following content:

```
apple
apricot
banana
apply
Apollo
```

Which of the following `grep` commands (using default BRE engine) would match all lines that start with "ap"?

Options :

6406534366719. ✖ `grep 'ap*' data.txt`

6406534366720. ✖ `grep '$ap' data.txt`

6406534366721. ✖ `grep 'ap$' data.txt`

6406534366722. ✔ `grep '^ap' data.txt`

✔ Official Answer Key: Option (D)

SC_036 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 237 Question Id : 6406531298631 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You have a file `input.txt` and you run the command: `grep -E '^(Task|Project) ID: [0-9]+' input.txt > results.txt` What is happening in this command?

Options :

6406534366723. ✔ Lines starting with either "Task" or "Project", followed by " ID: " and one or more digits, will have their standard output redirected to overwrite `results.txt`.

6406534366724. ✖ Lines containing "Task" or "Project" anywhere, followed by " ID: " and exactly one digit, will overwrite `results.txt`.

6406534366725. ✖ Lines starting with "Task" or "Project", followed by " ID: " and one or more digits, are appended to `results.txt`.

6406534366726. ✖ Only lines starting with "Task" and containing "Project ID: " and some numbers are saved to `results.txt`.

Sub-Section Number :	3
Sub-Section Id :	640653206811
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

SC_039 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 240 Question Id : 6406531298634 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the file `numbers.txt`:

```
123
45a
789
b01
2345
111
```

If you run the command `grep '[[:digit:]]\{3\}' numbers.txt` (using BRE), how many lines will be matched?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
4

Sub-Section Number :	5
Sub-Section Id :	640653206813
Question Shuffling Allowed :	No

✔ Official Answer Key (Numerical Value): 4

SC_049 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 250 Question Id : 640653470184 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7

Question Label : Multiple Choice Question

Select the regex to extract only the value of "id" from the below HTML file names as index.html

```
<html>
    <head>
        <title>My Site</title>
    </head>
    <body>
        <div class="container" id="container-secondary">
            <button id="load-preview-button">Load</button>
            <button id="play-video-button">Play</button>
        </div>
    </body>
</html>
```

Hint: `-o` option in grep prints only the matched regular expression.

```
grep "button" index.html
button
button
button
button
button
button
```

Options :

6406531562466. ✔ `grep 'id=".*"' -o index.html | cut -d'"' -f2`

6406531562467. ✖ `cat index.html | grep 'id=".*"' | cut -d'"' -f2`

6406531562468. ✖ `grep 'id=".*"' -o index.html`

6406531562469. ✖ `grep 'id="."' -o index.html | cut -d'"' -f1`

✔ Official Answer Key: Option (A)

SC_050 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 251 Question Id : 640653470185 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7

Question Label : Multiple Choice Question

Choose the regular expression that match with **ONLY** the five digit palindrome. Basic Regular Expression (BRE) is used.

Options :

6406531562470. ✖ `^.....$`

6406531562471. ✖ `^.{5}$`

6406531562472. ✔ `^\([0-9]\)\([0-9]\)[0-9]\2\1$`

6406531562473. ✖ `^\(.\)\(.\).\2\1$`

✔ **Official Answer Key: Option (C)**

SC_052 | [MSQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 253 Question Id : 640653470186 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8 Selectable Option : 0

Question Label : Multiple Select Question

```
(+91|0)?[6-9][[:digit:]]{9}
```

What will be the matched case(s) from the above regular expression. Assume that the regular expression is using Extended Regular Expression Engine (ERE). [MSQ]

Options :

6406531562474. ✖ Any ten digit mobile number

6406531562475. ✔ Any ten digit mobile number starting with 6, 7, 8 or 9

6406531562476. ✔ Any ten digit mobile number starting with 6, 7, 8 or 9 with 0 as prefix

6406531562477. ✔ Any ten digit mobile number starting with 6, 7, 8 or 9 with +91 as prefix

✔ Official Answer Key: Option (B), Option (C), Option (D)

SC_063 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 246 Question Id : 640653737534 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 9 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the regex to extract only the value of MAC address from the command `ifconfig` output as shown below.

Hint:

- A sample mac address is `00:15:5d:01:52:74`
- The field separator colon `:` separates 6 hexadecimal numbers

```
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1492
        inet 172.17.118.210  netmask 255.255.240.0  broadcast 172.17.127.255
        inet6 fe80::215:8dfa:fe06:5174  prefixlen 64  scopeid 0x20<link>
        ether 00:15:5d:01:52:74  txqueuelen 1000  (Ethernet)
        RX packets 1088  bytes 700086 (700.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 286  bytes 39079 (39.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 4  bytes 200 (200.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4  bytes 200 (200.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Hint:

- `-o` option in grep prints only the matched regular expression.
- `-E` option enables ERE

Options :

6406532468358. ✔ `grep -oE "([0-9A-Fa-f]{2}[:-]){5}([0-9A-Fa-f]{2})"`

6406532468359. ✖ `grep -o "([0-9A-Fa-f]{2}[:-]){5}([0-9A-Fa-f]{2})"`

6406532468360. ✔ `grep -o "\([0-9A-Fa-f]\{2\}[:-]\)\{5\}\([0-9A-Fa-f]\{2\}\)"`

✔ Official Answer Key: Option (A), Option (C)

SC_064 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 247 Question Id : 640653737541 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 9 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the command(s) that prints the files present in both the directory `dir1` and `dir2` in the current working directory.

Sample directory structure

```
$ ls -R
.:
dir1  dir2

./dir1:
file1  file2  file3  file4  file5  file6  file7

./dir2:
file1  file2  file3  file4  file5
```

Sample output

```
file1
file2
file3
file4
file5
```

Hints::

- `-f` flag in `grep` command will search for all the regex pattern in the file separated by line break
- `basename` command will return the filename from the path, example `basename /home/user/file1` will return `file1`

Options :

6406532468383. ✔ `ls dir2 > pat; ls dir1 | grep -f pat`

6406532468384. ✔ `ls dir1 > pat; ls dir2 | grep -f pat`

6406532468385. ✔ `for i in dir1/*; do [ -f "dir2/$(basename $i)" ] && echo $(basename $i); done`

✔ Official Answer Key: Option (A), Option (B), Option (C)

SC_068 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 251 Question Id : 640653737540 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

```
192.168.1.1 - - [22/Jan/2024:10:30:45 +0000] "GET /page1 HTTP/1.1" 200 1234
192.168.1.2 - - [22/Jan/2024:10:31:15 +0000] "GET /page2 HTTP/1.1" 404 5678
192.168.1.1 - - [22/Jan/2024:10:32:30 +0000] "POST /page1 HTTP/1.1" 200 9876
192.168.1.3 - - [22/Jan/2024:10:33:45 +0000] "GET /page3 HTTP/1.1" 200 4321
192.168.1.2 - - [22/Jan/2024:10:34:00 +0000] "GET /page1 HTTP/1.1" 200 8765
```

A sample web access log file of a web server is shown above. If the following command is run on the access log what will it return?

```
$ grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' web_access.log | sort | uniq -c
```

Hint:

Uniq with -c flag:

```
-c, --count
      prefix lines by the number of occurrences
```

Options :

6406532468379. ✖ Sort the log as per the first column entry

6406532468380. ✖ Sort the log as per the first column entry and show uniq entries in the log

6406532468381. ✖ Sort the log as per the first column entry and show uniq entries in the log with it's count in the output

6406532468382. ✔ sort the log as per first column entry and show the unique entries in the log with the output showing only the count of uniq entries in first column with the ip addresses as a first column from the log file

✔ Official Answer Key: Option (D)

SC_082 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 250 Question Id : 6406531113967 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Hexadecimal color codes are used for specifying colors for various elements in web pages. It consists of a hash (`#`), followed by three pairs of hexadecimal digits (0-9, A-F), e.g. `#1A2BCC` .

Which of the following regular expressions (extended) will correctly match hexadecimal color codes?

Options :

6406533774722. ✔ `^\#\([0-9A-Fa-f]\)\{6\}$`

6406533774723. ✖ `^\#\([0-9A-Fa-f]\)+$`

6406533774724. ✔ `^\#\([0-9A-Fa-f][0-9A-Fa-f]\)\{3\}$`

6406533774725. ✖ `\([0-9A-Fa-f]\)\{6\}`

Sub-Section Number :	6
Sub-Section Id :	640653168580
Question Shuffling Allowed :	No

Question Id : 6406531113969 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (251 to 255)

✔ **Official Answer Key: Option (A), Option (C)**

SC_092 | [MCQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 240 Question Id : 640653578004 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

Choose the most appropriate regex to match an email address. The regex is provided in Extended Regular Expression Engine (ERE).

Options :

6406531930165. ✖ `[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+.[A-Za-z]{2,}`

6406531930166.

✔ `\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\b`

6406531930167. ✖ `[A-Za-z0-9._%+-]+@*[A-Za-z0-9.-]+\.[A-Za-z]{2,}`

6406531930168. ✖ `\b[A-Za-z0-9._%+-]*@[A-Za-z0-9.-]*\.[A-Za-z]{2,}\b`

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

SC_096 | [MSQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 244 Question Id : 640653577997 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the regex to extract only the value of "email" from the below JSON file names as test.json.


```
[
  {
    "name": "Chadwick Cummings",
    "email": "nulla.dignissim.maecenas@hotmail.org",
    "address": "326-2072 Sagittis Road",
    "numberrange": 9,
    "alphanumeric": "ESM17JCJ7NR"
  },
  {
    "name": "Isaac Whitaker",
    "email": "vitae.semper.egestas@icloud.ca",
    "address": "589-9277 Vivamus St.",
    "numberrange": 3,
    "alphanumeric": "CTI05YDP7BX"
  },
  {
    "name": "Bethany Potter",
    "email": "enim.gravida@protonmail.com",
    "address": "P.O. Box 807, 2790 Ut, Ave",
    "numberrange": 7,
    "alphanumeric": "VOE77ZLE0OJ"
  }
]
```


Hint: `-o` option in grep prints only the matched regular expression.

```
grep -o "name" test.json
name
name
name
```

Options :

6406531930143. ✔ `grep "email" test.json | cut -d '"' -f 4`

6406531930144. ✔ `grep -o -E '"email": "[^"]+"' test.json | cut -d '"' -f 4`

6406531930145. ✖ `grep '"email":' test.json | cut -d ' ' -f 2`

6406531930146. ✖ `grep 'email: ' test.json | cut -d '"' -f 4`

✔ Official Answer Key: Option (A), Option (B)

SC_113 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 169 Question Id : 640653825161 Question Type : MSQ
Correct Marks : 7 Max. Selectable Options : 0
Question Label : Multiple Select Question

Select the **Regular Expression** to match the complete string of "email" (e.g. "email": "nulla.dignissim.maecenas@hotmail.org") from the below JSON file names as test.json.

```
[
        {
                "name": "Chadwick Cummings",
                "email": "nulla.dignissim.maecenas@hotmail.org",
                "address": "326-2072 Sagittis Road",
                "numberrange": 9,
                "alphanumeric": "ESM17JCJ7NR"
        },
        {
                "name": "Isaac Whitaker",
                "email": "vitae.semper.egestas@icloud.ca",
                "address": "589-9277 Vivamus St.",
                "numberrange": 3,
                "alphanumeric": "CTI05YDP7BX"
        },
        {
                "name": "Bethany Potter",
                "email": "enim.gravida@protonmail.com",
                "address": "P.O. Box 807, 2790 Ut, Ave",
                "numberrange": 7,
                "alphanumeric": "VOE77ZLE0OJ"
        }
]
```

Options :

6406532775138. ✔ `"email": "([^"]+)"`

6406532775139. ✖ `"email": \["(.+)"\]`

6406532775140. ✔
`"email"[[:space:]]*:[[:space:]]*"([^"]*)"`

6406532775141. ✖ `"email"[[:space:]]+:[[:space:]]+"([^"]*)"`

✔ Official Answer Key: Option (A), Option (C)

SC_115 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 171 Question Id : 640653825164 Question Type : MSQ

Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

You have a large log file, `server.log`, and you want the total count of the number of occurrences of the words "ERROR" and "WARNING" in the file. Which of the following command sequences will accomplish this?

Hint

- For command `grep`
  -c, --count
  Suppress normal output; instead print a count of matching lines for each input file. With the -v, --invert-match option (see below), count non-matching lines.
- In command `wc`
  -l, --lines
  print the newline counts

Options :

6406532775150. ✔ `grep -E "ERROR|WARNING" server.log | wc -l`

6406532775151. ✖ `cat server.log | grep "ERROR" | grep "WARNING" | wc -l`

6406532775152. ✖ `grep -c "ERROR" server.log && grep -c "WARNING" server.log`

6406532775153. ✔ `grep -E "ERROR|WARNING" server.log | wc -w`

✔ Official Answer Key: Option (A), Option (D)

SC_116 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 172 Question Id : 640653825174 Question Type : MSQ
Correct Marks : 8 Max. Selectable Options : 0
Question Label : Multiple Select Question

An IP (Internet Protocol) address is a unique identifier assigned to each device connected to a network that uses the Internet Protocol for communication. An IPv4 address is typically represented in the format `A.B.C.D`, where each part (A, B, C, D) is an integer between 0 and 255.

There are specific ranges of IP addresses reserved for private networks:

- **Class A**: 10.0.0.0 to 10.255.255.255
- **Class B**: 172.16.0.0 to 172.31.255.255
- **Class C**: 192.168.0.0 to 192.168.255.255

Given the following list of IP addresses:

1. 10.0.0.1
2. 10.255.255.254
3. 172.16.0.1
4. 172.31.255.254
5. 192.168.0.1
6. 192.168.255.254
7. 8.8.8.8
8. 169.254.1.1
9. 198.51.100.1
10. 203.0.113.1

Which of the following regular expressions (BRE) will match IP addresses from the Class A (10.x.x.x) and Class C (192.168.x.x) private network ranges?

Options :

6406532775187. ✖ `\(10\.\|192\.168\.\)[0-9]\{1,3\}\.[0-9]\{1,3\}`

6406532775188. ✔ `10\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\|192\.168\.[0-9]\{1,3\}\.[0-9]\{1,3\}`

6406532775189. ✖ `10\.\([0-9]\{1,3\}\.\)\{2\}[0-9]\{1,3\}\|192\.168\.\([0-9]\{1,3\}\.\)\{2\}`

6406532775190. ✔ `10\.\([0-9]\{1,3\}\.\)\{2\}[0-9]\{1,3\}\|192\.168\.\([0-9]\{1,3\}\.\)\{2\}`

Sub-Section Number :	7
Sub-Section Id :	640653122776
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B), Option (D)

SC_125 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 137 Question Id : 6406531527872 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You run the command:

```
1 grep -E '^[A-Za-z]{3}[0-9]{2}$' ids.txt
```

What will be matched?

Options :

6406535115759. ✖ Lines starting with 3 letters and containing 2 digits anywhere.

6406535115760. ✔ Lines containing exactly 3 letters followed by exactly 2 digits.

6406535115761. ✖ Lines ending with 2 digits.

6406535115762. ✖ Any line containing at least 3 letters and 2 digits.

✔ Official Answer Key: Option (B)

SC_131 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 143 Question Id : 6406531527877 Question Type : SA
Correct Marks : 4
Question Label : Short Answer Question

Consider the file `ips.txt` containing the following IP addresses:

```
10.0.0.1
172.16.5.4
192.168.1.1
8.8.8.8
127.0.0.1
15.12.34.56
130.25.36.48
1.2.3.4
100.50.25.75
200.100.50.25
```

You run the following command:

```
grep -E '^([1-9]|[1-9][0-9]|1[0-1][0-9]|12[0-6])\.' ips.txt | wc -l
```

What will be the output of this command?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
5

Sub-Section Number :	7
Sub-Section Id :	640653244935
Question Shuffling Allowed :	No

Question Id : 6406531527878 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (144 to 145)

✔ **Official Answer Key (Numerical Value): 5**

SC_138 | [MCQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025 (+3 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 127 Question Id : 6406531525141 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You run the command:

```
1 grep -E '^[A-Za-z]{3}[0-9]{2}$' ids.txt
```

What will be matched?

Options :

6406535108122. ✖ Lines starting with 3 letters and containing 2 digits anywhere.

6406535108123. ✔ Lines containing exactly 3 letters followed by exactly 2 digits.

6406535108124. ✖ Lines ending with 2 digits.

6406535108125. ✖ Any line containing at least 3 letters and 2 digits.

✔ **Official Answer Key: Option (B)**

SC_141 | [SA] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025 (+5 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 133 Question Id : 6406531525146 Question Type : SA
Correct Marks : 4
Question Label : Short Answer Question

Consider the file `ips.txt` containing the following IP addresses:

```
10.0.0.1
172.16.5.4
192.168.1.1
8.8.8.8
127.0.0.1
15.12.34.56
130.25.36.48
1.2.3.4
100.50.25.75
200.100.50.25
```

You run the following command:

```
grep -E '^([1-9]|[1-9][0-9]|1[0-1][0-9]|12[0-6])\.' ips.txt | wc -l
```

What will be the output of this command?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
5

Sub-Section Number : 7
Sub-Section Id : 640653244040
Question Shuffling Allowed : No

Question Id : 6406531525147 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (134 to 135)

✔ Official Answer Key (Numerical Value): 5

SC_142 | [MCQ] | Marks: +3.0 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025 (+1 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 175 Question Id : 6406531527081 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You run the command:

```
1 grep -E '^[A-Za-z]{3}[0-9]{2}$' ids.txt
```

What will be matched?

Options :

6406535113573. ✖ Lines starting with 3 letters and containing 2 digits anywhere.

6406535113574. ✔ Lines containing exactly 3 letters followed by exactly 2 digits.

6406535113575. ✖ Lines ending with 2 digits.

6406535113576. ✖ Any line containing at least 3 letters and 2 digits.

✔ Official Answer Key: Option (B)

SC_144 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 224 Question Id : 640653668672 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

Select a command from the given options which will find and print **a number** hidden in the text comprising of [a-zA-Z.]. Note that there is only one numeric character while rest of the characters are either alphabets or punctuation. Command which prints only the number and not text.

```
$ cat hidden_number.txt
Loremipsumdolorsitame
t,consecteturadipisci
ngelit,seddoeiusmodte
mporincididuntutlabor
eetdoloremagnaaliqua.
Condimentumlaciniaqui
sveleros.Sitametmauri
scommodoquisimperdiet
massatinciduntnunc.Le
ctusurnaduisconvallis
convallistellusidinte
rdumvelitlaoreet.Ferm
entumetsollicitudinac
orci.Aarcucursusvitae
conguemaurisrhoncusae
neanvel.Acfeugiatsed1
ectusvestibulummattis
ullamcorper.Sedadipis
cingdiamdonecadipisci
ngtristiquerisus.Volu
$ grep --help
Usage: grep [OPTION]... PATTERNS [FILE]...
Search for PATTERNS in each FILE.
Example: grep -i 'hello world' menu.h main.c
PATTERNS can contain multiple patterns separated by newlines.

...
  -o, --only-matching       show only non-empty parts of lines that match
...
```

Options :

6406532240222. ✖ `grep -v '[a-zA-Z,.]' hidden_number.txt`

6406532240223. ✖ `grep '[a-zA-Z,.]' hidden_number.txt`

6406532240224.

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

SC_145 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 225 Question Id : 640653668681 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

ISO 8601 date-time formats in UTC are shown below. From the options select the correct extended regular expression that will match with all of the given formats.

```
# UTC date-time formats

2022-05-06T13:00:56+00:00
2022-05-06T13:00:56Z
20220506T130056Z
```

Options :

6406532240243. ✖ `^[0-9]{4}-*[0-9]{2}-*[0-9]{2}T[0-9]{2}:*[0-9]{2}:*[0-9]{2}(Z|\+00:00)$`

6406532240244. ✖ `^[0-9]{4}-+[0-9]{2}-+[0-9]{2}T[0-9]{2}:+[0-9]{2}:+[0-9]{2}(Z|\+00:00)$`

6406532240245. ✔ `^[0-9]{4}-?[0-9]{2}-?[0-9]{2}T[0-9]{2}:?[0-9]{2}:?[0-9]{2}(Z|\+00:00)$`

6406532240246. ✖ `^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}(Z|\+00:00)$`

✔ Official Answer Key: Option (C)

SC_162 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 139 Question Id : 640653995567 Question Type : MSQ
Correct Marks : 7 Max. Selectable Options : 0
Question Label : Multiple Select Question
Select the **Regular Expression** to match the complete string of "address" (e.g. `"address": "326-2072 Sagittis Road"`) from the below JSON file named `addresses.json`.

```
[
        {
                "name": "Chadwick Cummings",
                "email": "nulla.dignissim.maecenas@hotmail.org",
                "address": "326-2072 Sagittis Road",
                "numberrange": 9,
                "alphanumeric": "ESM17JCJ7NR"
        },
        {
                "name": "Isaac Whitaker",
                "email": "vitae.semper.egestas@icloud.ca",
                "address": "589-9277 Vivamus St.",
                "numberrange": 3,
                "alphanumeric": "CTI05YDP7BX"
        },
        {
                "name": "Bethany Potter",
                "email": "enim.gravida@protonmail.com",
                "address": "P.O. Box 807, 2790 Ut, Ave",
                "numberrange": 7,
                "alphanumeric": "VOE77ZLE0OJ"
        },

]
```

Options :

6406533361017. ✔ `"address": "([^"]+)"`

6406533361018. ✖ `"address": \["(.+)"\]`

6406533361019. ✔ `"address"[[:space:]]*:[[:space:]]*"([^"]*)"`

6406533361020. ✖ `"address"[[:space:]]+:[[:space:]]+"([^"]*)"`

Sub-Section Number :	5
Sub-Section Id :	640653146823
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (C)

SC_164 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 141 Question Id : 640653995570 Question Type : MSQ

Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

You have a large log file, `server.log`, and you want to count the number of occurrences of the word "ERROR" in the file. Which of the following command sequences will accomplish this?

Hint

- For command `grep`

  -c, --count

  Suppress normal output; instead print a count of matching lines for each input file. With the -v, --invert-match option (see below), count non-matching lines.
- In command `wc`

  -l, --lines

  print the newline counts

Options :

6406533361029. ✔ `grep "ERROR" server.log | wc -l`

6406533361030. ✖ `echo server.log | grep "ERROR" | wc -l`

6406533361031. ✖ `wc -l server.log | grep "ERROR"`

6406533361032. ✔ `grep -c "ERROR" server.log`

✔ **Official Answer Key: Option (A), Option (D)**

SC_170 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 147 Question Id : 640653995580 Question Type : MSQ

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

A MAC (Media Access Control) address is a unique identifier assigned to network interfaces for communications on the physical network segment. A MAC address is typically represented in the format `00:1A:2B:3C:4D:5E`, where each pair of characters is a hexadecimal digit (0-9, A-F).

Which of the following regular expressions (Extended Regular Expression) will match a MAC address in the format `00:1A:2B:3C:4D:5E`?

Options :

6406533361066. ✔ `\([0-9A-Fa-f]\{2\}:\)\{5\}[0-9A-Fa-f]\{2\}`

6406533361067. ✔ `[0-9A-Fa-f]\{2\}:[0-9A-Fa-f]\{2\}:[0-9A-Fa-f]\{2\}:[0-9A-Fa-f]\{2\}:[0-9A-Fa-f]\{2\}:[0-9A-Fa-f]\{2\}:[0-9A-Fa-f]\{2\}`

6406533361068. ✖ `[0-9A-F]\{2\}:[0-9A-F]\{2\}:[0-9A-F]\{2\}:[0-9A-F]\{2\}:[0-9A-F]\{2\}:[0-9A-F]\{2\}`

6406533361069. ✖ `[0-9A-Fa-f]\{2\}-[0-9A-Fa-f]\{2\}-[0-9A-Fa-f]\{2\}-[0-9A-Fa-f]\{2\}-[0-9A-Fa-f]\{2\}-[0-9A-Fa-f]\{2\}`

6406533361070. ✖ `\([0-9A-Fa-f]\{2\}:\)\{6\}`

Sub-Section Number :	7
Sub-Section Id :	640653146825
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B)

SC_176 | [MSQ] | Marks: +3.0 | *System commands*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 4 Question Id : 6406532159059 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

A file named `server.log` contains the following four raw text lines:

```
system_error_404
system_fail_500
system_error_4040
error_404
```

An administrator uses different variations of regular expressions to extract specific log entries.

Which of the following statements is/are **TRUE** regarding the execution and matching behavior of these commands?

Options :

6406537023184. ✖ The command `grep "error_(404|500)" server.log` will successfully match and output the line `system_error_404`.

6406537023185. ✔ To make standard `grep` match the same lines as `egrep "error_(404|500)" server.log`, the pattern can be rewritten as: `grep "error_\(404\|500\)" server.log`.

6406537023186. ✔ The command `egrep "error_404\b" server.log` will match `system_error_404` and `error_404`, but will exclude `system_error_4040`.

6406537023187. ✖ The command `grep "error_404\b" server.log` will output nothing because word boundary assertions (`\b`) are only supported in Extended Regular Expressions (`egrep`).

✔ Official Answer Key: Option (B), Option (C)

SC_177 | [MSQ] | Marks: +3.0 | *System commands*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.3 Pattern Searching with Grep & Regular Expressions

Question Number : 5 Question Id : 6406532159060 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

You have created a file named `files.txt` that contains the following filenames:

```
file1.txt
file2.txt
file10.txt
file20.txt
```

You want to extract only the lines that contain `file1.txt` and `file2.txt` (but not `file10.txt` or `file20.txt`).

Which of the following grep commands will correctly achieve this?

Options :

6406537023188. ✔ `grep -E '^file(1|2)\.txt$' files.txt`

6406537023189. ✔ `grep -E '^file[12]\.txt$' files.txt`

6406537023190. ✖ `grep -E '^file1|^file2' files.txt`

6406537023191. ✖ `grep -E '^file1|2\.txt$' files.txt`

Sub-Section Number :	4
Sub-Section Id :	640653396306
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B)

• 3.2 Finding Files & Path Searches (9 Questions)

SC_016 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.2 Finding Files & Path Searches

Question Number : 269 Question Id : 640653345422 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

```
$ pwd
/home/runner
$ ls -l
total 8
-rw-r--r-- 1 runner runner 105 May 19 10:52 main.sh
-rw-r--r-- 1 runner runner  62 Nov 11  2021 replit.nix

$ cd /
$ cd home/runner
$ COMMAND1 # Find correct command for this.
$ long_listing
total 8
-rw-r--r-- 1 runner runner 105 May 19 10:52 main.sh
-rw-r--r-- 1 runner runner  62 Nov 11  2021 replit.nix
```

There is a command which is missing in the command sequence above, it is marked as `COMMAND1`. Analyze the outputs of the commands and select the command from the below options that is the correct replacement for `COMMAND1`.

Options :

6406531148855. ✖ `long_listing="ls -l"`

6406531148856. ✔ `alias long_listing="ls -l"`

6406531148857. ✖ `alias long_listing "ls -l"`

6406531148858. ✖ `apropos long_listing "ls -l"`

6406531148859. ✖ None of these.

✔ Official Answer Key: Option (B)

SC_057 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.2 Finding Files & Path Searches

Question Number : 240 Question Id : 640653737527 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

Ajit is writing an essay in English. He realizes that he has used both the American English and British English spellings for the same words in his essay. He wants to find all such occurrences for the words "color/colour" and "program/programme". Help him by selecting the correct regular expression which will capture the British version as well as American version of the words from the text. Make sure It only collects the words which has correct spellings in either of the format.

Note: It is an Extended Regular Expression (ERE)

Options :

6406532468329. ✖ `colo?r`

6406532468330. ✖ `program???`

6406532468331. ✔ `colou?r`

6406532468332. ✔ `program(me)?`

✔ **Official Answer Key: Option (C), Option (D)**

SC_062 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.2 Finding Files & Path Searches

Question Number : 245 Question Id : 640653737539 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

Following bash script is run to find if a given file has a 'txt' extension. Which of the following option(s) will achieve the same?

```
#!/bin/bash

echo "Enter a filename: "
read filename

if [[ "$filename" =~ \.txt$ ]]; then
    echo "Text file"
else
    echo "Not a text file"
fi
```

Options :

6406532468373. ✔ `extension="txt"; [ "${filename##*.}" == "$extension" ] && echo "Text file" || echo "Not a Text file"`

6406532468374. ✖ `extension="txt"; [ "${filename%%*.}" == "$extension" ] && echo "Text file" || echo "Not a Text file"`

6406532468375. ✖ `` `extension="txt"; [ "${filename%%*.}" == "$extension" ] || echo "Text file" && echo "Not a Text file" ``

6406532468376. ✖ `extension="txt"; [ "${filename##*.}" == "$extension" ] || echo "Text file" && echo "Not a Text file"`

6406532468377. ✔ `[[ "$filename" =~ \.txt$ ]] && echo "Text file" || echo "Not a text file"`

6406532468378. ✖ `[[ "$filename" =~ \.txt$ ]] || echo "Text file" && echo "Not a text file"`

✔ Official Answer Key: Option (A), Option (E)

SC_126 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.2 Finding Files & Path Searches

Question Number : 138 Question Id : 6406531527873 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

File: `/home/user/files/files.txt`

Contents:

```
report1.txt
report2.doc
summary.txt
notes.csv
backup.txt
report123.txt
report.txt
```

Find all lines that start with `report` and end with `.txt`.

Which command is correct?

Options :

6406535115763. ✖ `grep "report\.*txt" /home/user/files/files.txt`

6406535115764. ✔ `grep "^report.*\.txt$" /home/user/files/files.txt`

6406535115765. ✖ `egrep "^report*.txt" /home/user/files/files.txt`

6406535115766. ✖ `grep "^report.txt$" /home/user/files/files.txt`

Sub-Section Number :	3
Sub-Section Id :	640653244931
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

SC_128 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.2 Finding Files & Path Searches

Question Number : 140 Question Id : 6406531527871 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

File: `/home/user/numbers/numbers.txt`

Contents:

```
ID: 102
ID: 215
ID: abc
ID: 330
ID: xyz
ID: 22A
ID: 2222
ID: A111
```

Find lines where ID is exactly 3 digits.

Which command is correct?

Options :

6406535115755. ✔ `egrep "ID: [0-9]{3}$" /home/user/numbers/numbers.txt`

6406535115756. ✖ `egrep "ID: [0-9]{2,3}" /home/user/numbers/numbers.txt`

6406535115757. ✖ `egrep "ID: [0-9]{3,}" /home/user/numbers/numbers.txt`

6406535115758. ✔ `grep "^ID: [0-9][0-9][0-9]$" /home/user/numbers/numbers.txt`

✔ Official Answer Key: Option (A), Option (D)

SC_139 | [MCQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025 (+5 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.2 Finding Files & Path Searches

Question Number : 128 Question Id : 6406531525142 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

File: `/home/user/files/files.txt`

Contents:

```
report1.txt
report2.doc
summary.txt
notes.csv
backup.txt
report123.txt
report.txt
```

Find all lines that start with `report` and end with `.txt`.

Which command is correct?

Options :

6406535108126. ✖ `grep "report\.*txt" /home/user/files/files.txt`

6406535108127. ✔ `grep "^report.*\.txt$" /home/user/files/files.txt`

6406535108128. ✖ `egrep "^report*.txt" /home/user/files/files.txt`

6406535108129. ✖ `grep "^report.txt$" /home/user/files/files.txt`

Sub-Section Number :	3
Sub-Section Id :	640653244036
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

SC_140 | [MSQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025 (+5 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.2 Finding Files & Path Searches

Question Number : 130 Question Id : 6406531525140 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

File: `/home/user/numbers/numbers.txt`

Contents:

```
ID: 102
ID: 215
ID: abc
ID: 330
ID: xyz
ID: 22A
ID: 2222
ID: A111
```

Find lines where ID is exactly 3 digits.

Which command is correct?

Options :

6406535108118. ✔ `egrep "ID: [0-9]{3}$" /home/user/numbers/numbers.txt`

6406535108119. ✖ `egrep "ID: [0-9]{2,3}" /home/user/numbers/numbers.txt`

6406535108120. ✖ `egrep "ID: [0-9]{3,}" /home/user/numbers/numbers.txt`

6406535108121. ✔ `grep "^ID: [0-9][0-9][0-9]$" /home/user/numbers/numbers.txt`

✔ Official Answer Key: Option (A), Option (D)

SC_147 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.2 Finding Files & Path Searches

Question Number : 227 Question Id : 640653668684 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

What the following `for` loop will print at the end of the execution?

```
for file in `find . -maxdepth 1 -name '*.txt'`; do
  echo $(basename $file)
done
```

Options :

6406532240257. ✔ Prints the file names of all the '.txt' files in the current directory non recursively.

6406532240258. ✖ Prints the file names of all the '.txt' files in the current directory recursively.

6406532240259. ✖ Prints the file names of all the '.txt' files in the current directory and in immediate subdirectories of current working directory non-recursively.

Sub-Section Number :	3
Sub-Section Id :	64065395214
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

SC_186 | [MSQ] | Marks: +3.0 | *System commands*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.2 Finding Files & Path Searches

Question Number : 14 Question Id : 6406532159061 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question
File: `/etc/db/cluster.conf`

Note: Answer this question only based on the file contents given below.

`cluster.conf` Contents:

```
node_port=8080
node_port=80
node_port=8085
NODE_PORT=8080
node_port=9000
node_port=808
mynode_port=8080
node_port=80802
node_port=80a0
node_port=8081
```

You want to find all lines that **start exactly with** `node_port=` **(lowercase only)** and **end with a 4-digit port number that specifically begins with** `80` (i.e., ports in the range `8000` to `8099`).

Which of the following commands will correctly print all and only the required lines?
Options :

6406537023192. ✖ `grep "^node_port=80.*$" /etc/db/cluster.conf`

6406537023193. ✔ `grep -E "^node_port=80[0-9]{2}$" /etc/db/cluster.conf`

6406537023194. ✖ `grep -i "^node_port=80[0-9][0-9]$" /etc/db/cluster.conf`

6406537023195. ✔ `grep "^node_port=80[0-9][0-9]$" /etc/db/cluster.conf`

Sub-Section Number : 6
Sub-Section Id : 640653396308
Question Shuffling Allowed : Yes

✔ Official Answer Key: Option (B), Option (D)

• 3.1 Text Processing Filters (9 Questions)

SC_044 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.1 Text Processing Filters

Question Number : 245 Question Id : 640653470189 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

Choose the line that modified the contents inside the file `backup.sh`

```
cat backup.sh | cut -d " " -f 1
echo -n >> backup.sh
echo hello world > backup.sh
cp backup.sh backup.2023.sh
```

Options :

6406531562483. ✖ `cat backup.sh | cut -d " " -f 1`

6406531562484. ✖ `echo -n >> backup.sh`

6406531562485. ✔ `echo hello world > backup.sh`

6406531562486. ✖ `cp backup.sh backup.2023.sh`

✔ **Official Answer Key: Option (C)**

SC_074 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.1 Text Processing Filters

Question Number : 242 Question Id : 6406531113960 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

What will be the output of the following commands?

```
mkdir example; cd example
touch .file-{1..3}{a..c}; ls | wc -l
```

Options :

6406533774694. ✖ 9

6406533774695. ✖ 6

6406533774696. ✖ 3

6406533774697. ✔ 0

✔ **Official Answer Key: Option (D)**

SC_091 | [MCQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.1 Text Processing Filters

Question Number : 239 Question Id : 640653578002 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

Following lines are a part of a shell script for job submission to a server. Assume that the PBS_JOBID stores `123456.pbs` value and $HOME stores `/user` value. What would be the value of `tempdir` variable.

```
tpdir=`echo $PBS_JOBID | cut -f 1 -d .`
tempdir=$HOME/scratch/job$tpdir
mkdir -p $tempdir
cd $tempdir
```

Options :

6406531930160. ✖ `/user/scratch/jobpbs`

6406531930161. ✖ `job123456`

6406531930162. ✔ `/user/scratch/job123456`

6406531930163. ✖ `/scratch/jobpbs`

✔ Official Answer Key: Option (C)

SC_093 | [MCQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.1 Text Processing Filters

Question Number : 241 Question Id : 640653578005 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

```
$ tr --help
Usage: tr [OPTION]... SET1 [SET2]
Translate, squeeze, and/or delete characters from standard
input,
writing to standard output.

  -c, -C, --complement    use the complement of SET1
  -d, --delete            delete characters in SET1, do
not translate
  -s, --squeeze-repeats   replace each sequence of a
repeated character
                            that is listed in the last
specified SET,
                            with a single occurrence of
that character
  -t, --truncate-set1     first truncate SET1 to length of
SET2
      --help     display this help and exit
      --version  output version information and exit

SETs are specified as strings of characters.  Most
represent themselves.
Interpreted sequences are:
```

✔ Official Answer Key: Option (C)

SC_100 | [SA] | Marks: +3.0 | *Quiz1 May23*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.1 Text Processing Filters

Question Number : 248 Question Id : 640653578000 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7

Question Label : Short Answer Question

`man cut` partial output

```
CUT(1)                                                    User
Commands
CUT(1)

NAME
       cut - remove sections from each line of files

SYNOPSIS
       cut OPTION... [FILE]...

DESCRIPTION
       Print selected parts of lines from each FILE to
standard output.

       With no FILE, or when FILE is -, read standard
input.

       Mandatory arguments to long options are mandatory
for short options too.

      -b, --bytes=LIST
             select only these bytes

      -c, --characters=LIST
             select only these characters

      -d, --delimiter=DELIM
             use DELIM instead of TAB for field delimiter

       -f, --fields=LIST
              select  only  these  fields;   also print
any line that contains no delimiter character, unless the
-s
              option is specified
```

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

SC_124 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.1 Text Processing Filters

Question Number : 136 Question Id : 6406531527870 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You want to:

- Combine the contents of two files: `intro.txt` and `summary.txt`
- Sort all lines alphabetically
- Save the final sorted output to a file named `full_report.txt`, **overwriting** any existing content

Which of the following commands correctly performs **all** of these steps?

Options :

6406535115751. ✖ `cat intro.txt | cat summary.txt | sort > full_report.txt`

6406535115752. ✖ `cat intro.txt summary.txt | sort >> full_report.txt`

6406535115753. ✖ `cat intro.txt > full_report.txt && cat summary.txt | sort`

6406535115754. ✔ `cat intro.txt summary.txt | sort > full_report.txt`

✔ Official Answer Key: Option (D)

SC_151 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.1 Text Processing Filters

Question Number : 231 Question Id : 640653668671 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 9 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select command(s)/script(s) from the options which will extract the inner rectangle made of 'x' from the pattern shown below which is stored in file `bounded_rectangle.txt`.

Note: Boundary spaces are not considered but the proportionality, scale, and number of characters of inner rectangle should be the same.

Sample Input: bounded_rectangle.txt

```
**************************
*                        *
*                        *
* xxxxxxxxxxxxxxxxxxxxxx *
* x                    x *
* x                    x *
* x                    x *
* x                    x *
* xxxxxxxxxxxxxxxxxxxxxx *
*                        *
*                        *
**************************
```

Sample Output

```
xxxxxxxxxxxxxxxxxxxxxx
x                    x
x                    x
x                    x
x                    x
xxxxxxxxxxxxxxxxxxxxxx
```

Hint:

```
$ echo -e '123\n12'
123
12

$ echo -e '123\n12' | rev
321
21
```

Options :

6406532240217. ✔ `cat bounded_rectangle.txt | grep x | cut -d'*' -f2`

6406532240218. ✔ `cat bounded_rectangle.txt | cut -d' ' -f2- | rev | cut -d' ' -f2- | grep x`

✔ Official Answer Key: Option (A), Option (B)

SC_156 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.1 Text Processing Filters

Question Number : 236 Question Id : 640653668680 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7

Question Label : Short Answer Question

Below is the file hierarchy structure provided to answer the question.

```
.
├── classA
│   └── attendance
│       ├── feb.csv
│       ├── jan.csv
│       └── mar.csv
├── classB
│   └── attendance
│       ├── feb.csv
│       ├── jan.csv
│       └── mar.csv
├── classC
│   └── attendance
│       ├── feb.csv
│       ├── jan.csv
│       └── mar.csv
└── script.sh

6 directories, 10 files
```

Following script is executed on the directory `classA` (`dir` variable). For what value (`num` variable) will the script output `true` upon execution of the script.

```
#!/bin/bash
dir=$1
num=$2
[[ $num -eq `ls $dir | wc -l` ]] && echo true || echo false
```

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

✔ ***Official Answer Key:*** *[Refer to detailed question paper]*

SC_157 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.1 Text Processing Filters

Question Number : 237 Question Id : 640653668679 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 8

Question Label : Short Answer Question

What will be the output of the following command?

```
cat /etc/passwd |tr ':' ','|head -30|tail -10|sort -k4,4|cut -d, -f3|head -1
```

Following is the relevant part of `/etc/passwd` file on which the command is run (i.e. output of the `cat /etc/passwd |tr ':' ','|head -30|tail -10` command).

```
syslog:x:102:106::/home/syslog:/usr/sbin/nologin
messagebus:x:103:107::/nonexistent:/usr/sbin/nologin
_apt:x:104:65534::/nonexistent:/usr/sbin/nologin
lxd:x:105:65534::/var/lib/lxd/:/bin/false
uuidd:x:106:110::/run/uuidd:/usr/sbin/nologin
dnsmasq:x:107:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
landscape:x:108:112::/var/lib/landscape:/usr/sbin/nologin
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
pollinate:x:110:1::/var/cache/pollinate:/bin/false
systemcommands:x:1000:1000:,,,:/home/systemcommands:/bin/bash
```

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

104

✔ Official Answer Key (Numerical Value): 104

• 3.4 Stream Editing with Sed (5 Questions)

SC_060 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.4 Stream Editing with Sed

Question Number : 243 Question Id : 640653737530 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

```
$ sleep 200 &
$ ps
    PID TTY          TIME CMD
     10 pts/0    00:00:02 bash
    344 pts/0    00:00:00 sleep
    345 pts/0    00:00:00 ps
```

Select the command(s) from the options below which will suspend the `sleep` process started as shown in the above code block.

```
$ man kill
KILL(1)                          User Commands
KILL(1)

NAME
       kill - send a signal to a process

SYNOPSIS
       kill [options] <pid> [...]

DESCRIPTION
       The  default  signal  for kill is TERM.  Use -l or -L to list
available signals.
       Particularly useful signals include HUP, INT, KILL, STOP, CONT, and
0.   Alter-
       nate  signals  may  be specified in three ways: -9, -SIGKILL or -
KILL.  Negative
       PID values may be used to choose whole process groups; see the PGID
column in ps
       command  output.   A PID of -1 is special; it indicates all processes
except the
       kill process itself and init.
OPTIONS
       <pid> [...]
              Send signal to every <pid> listed.
```

✔ Official Answer Key: Option (C)

SC_099 | [MSQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.4 Stream Editing with Sed

Question Number : 247 Question Id : 640653578007 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

```
#!/bin/bash

directory="mydirectory"
zip_dest="myarchive.zip"
read password
zip -r -P $password $zip_dest $directory
```

Hint:

```
$ man zip
ZIP(1)                      General Commands Manual
ZIP(1)

NAME
       zip - package and compress (archive) files

SYNOPSIS
       zip  [-aABcdDeEfFghjklLmoqrRSTuvVwXyz!@$] [--
longoption ...]  [-b path]
       [-n suffixes] [-t date] [-tt date] [zipfile [file
...]]  [-xi list]

...

DESCRIPTION
       zip  is  a compression and file packaging utility
for Unix, VMS, MSDOS,
       OS/2, Windows 9x/NT/XP, Minix, Atari, Macintosh,
Amiga, and Acorn  RISC
       OS.   It  is analogous to a combination of the Unix
commands tar(1) and
       compress(1) and is compatible with PKZIP (Phil
Katz's  ZIP  for  MSDOS
       systems).
```

✔ Official Answer Key: Option (A)

SC_143 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.4 Stream Editing with Sed

Question Number : 223 Question Id : 640653668670 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

By using the information provided below, select a command/script from the options which prints a line with the process ID (PID) for the command `daily-backup`. The command is **only** (exclusively) started by the current user.

A sample output from `ps` command.

```
$ ps x
    PID TTY      STAT   TIME COMMAND
   1168 ?        Ss     0:00 /bin/sh -c cd /home/vidya/quicklinks; python3 -
m http.server 7777
   2331 ?        S      0:00 /usr/bin/bash
/home/vidya/server/deployment_script.sh
   2381 ?        S      0:00 /usr/bin/bash
/home/vidya/server/deployment_script.sh
   5293 ?        S      0:00 (sd-pam)
   8293 ?        S      0:00 /usr/bin/bash
   1381 ?        S      0:00 /home/vidya/.local/bin/daily-backup
   1361 ?        S      0:00 /home/vidya/.local/bin/daily-backuper.sh

$ ps
    PID TTY          TIME COMMAND
   8293 pts/32       0:00 /usr/bin/bash
   8973 pts/32       0:00 ps

$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root       96641  0.0  0.0      0     0 ?        I<   16:04   0:00
[kworker/u17:1-i915_flip]
root       96656  0.0  0.0      0     0 ?        I    16:04   0:00
[kworker/3:1-events]
root       96827  0.0  0.0      0     0 ?        I    16:04   0:00
[kworker/1:1-events]
vidya   96844  0.3  1.1 1188997100 90548 ?    Sl   16:05   0:00
/opt/google/chrome/chrome
vidya   96919  0.0  0.1  21608  8216 pts/21   S    16:05   0:00 /usr/bin/zsh
vidya   96920  0.0  0.1  21640  8224 pts/21   S    16:05   0:00 /usr/bin/zsh
vidya   96921  0.0  0.0  13716  3328 pts/21   R+   16:05   0:00 ps aux
```

Options :

6406532240213. ✔ `ps x | grep '\b/home/vidya/.local/bin/daily-backup\b'`

6406532240214. ✖ `ps | grep daily-backup`

6406532240215.

✔ Official Answer Key: Option (A)

SC_146 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.4 Stream Editing with Sed

Question Number : 226 Question Id : 640653668682 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

Following code block shows the commands ran in terminal. What will be the output of the last command?

```
$ ps
    PID TTY          TIME CMD
  64754 pts/3    00:00:00 bash
  65657 pts/3    00:00:00 sleep
  65671 pts/3    00:00:00 sleep
  65718 pts/3    00:00:00 python3
  65774 pts/3    00:00:00 ps

$ ps | grep "$$"
```

Options :

6406532240247. ✖ `PID TTY TIME CMD`

6406532240248. ✔ `64754 pts/3 00:00:00 bash`

6406532240249. ✖ `65718 pts/3 00:00:00 python3`

6406532240250. ✖ `65774 pts/3 00:00:00 ps`

✔ Official Answer Key: Option (B)

SC_188 | [MSQ] | Marks: +3.0 | *System commands*

Category: Week 3: Text Processing Filters, Grep/Regex, Sed & Awk › 3.4 Stream Editing with Sed

Question Number : 16 Question Id : 6406532159067 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
You want to perform the following operations:

1. Create a new nested directory path named `utils/helpers` directly inside `src`.
2. Create an empty file named `string_utils.py` inside that new `src/utils/helpers` directory.
3. Inside the `test` directory, create nine new empty test files named `test_1.py`, `test_2.py`, ... up to `test_9.py`.

Which of the following command sets will perform all the above operations correctly?
Options :

```
mkdir -p src/utils/helpers
cd src/utils/helpers
touch string_utils.py
cd ../../../test
touch test_{1..9}.py
```

6406537023212. ✔

```
mkdir src/utils/helpers
touch src/utils/helpers/string_utils.py
touch test/test_{1..9}.py
```

6406537023213. ✖

6406537023214. ✔

```
mkdir -p src/utils/helpers
touch src/utils/helpers/string_utils.py
touch test/{test_1.py,test_2.py,test_3.py,test_4.py,test_5.py,test_6.py,test_7.py,test_8.py,test_9.py}
```

```
mkdir -p src/utils/helpers
touch src/utils/helpers/string_utils.py
touch test/test_1..9.py
```

6406537023215. ✖

✔ Official Answer Key: Option (A), Option (C)

Week 4: Shell Scripting, Environment Variables & Process Management (40 Questions)

• 4.2 Shell Scripting: Conditionals & Test Operators (23 Questions)

SC_006 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 259 Question Id : 640653345408 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Select Question

Initially, four files named `a`, `b`, `c` and `d` are present in the current working directory, every file has some text in it. Select the file(s) whose contents are not modified at the end of executing the below set of commands in the same sequence.

Hint: Description of option with `echo` command.

-n: do not output the trailing newline

```
touch a
echo -n ""> b  #There is no space between quotes.
echo -n "">> c  #There is no space between quotes.
cat a > d
```

Options :

6406531148805. ✔ a

6406531148806. ✖ b

6406531148807. ✔ c

6406531148808. ✖ d

✔ Official Answer Key: Option (A), Option (C)

SC_009 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 262 Question Id : 640653345410 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

```
! (ls global_config || ls local_config) && touch default_config
```

The file `default_config` does not exist in the current working directory. All the statements below are with respect to the current working directory.

After running the above command the file `default_config` will be created iff,

Note: `&&` and `||` have same precedence.

Options :

6406531148810. ✖ The files `global_config` and `local_config` both exist.

6406531148811. ✖ The file `local_config` exists but `global_config` does not exist.

6406531148812. ✖ The file `global_config` exists but `local_config` does not exist.

6406531148813. ✔ The files `global_config` and `local_config` both do not exist.

✔ Official Answer Key: Option (D)

SC_030 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 231 Question Id : 6406531298625 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the following editing task: **Navigate to the very first line of the currently open file, remove that entire line, then save all changes, and finally exit the editor.**

Which of the following options correctly lists the command sequences to perform this task first in the `vi` editor (starting from command mode after opening `filename.txt`), and then in the `nano` editor (after opening `filename.txt`)?

Options :

- `vi filename.txt` (then in vi command mode):
  - `1G`
  - `dd`
  - `:wq`
- `nano filename.txt` (then in nano):
  - `Alt+\`
  - `Ctrl+K`
  - `Ctrl+O`, Enter
  - `Ctrl+X`

6406534366699. ✔

- `vi filename.txt` (then in vi command mode):
  - `:1`
  - `D`
  - `:x`
- `nano filename.txt` (then in nano):
  - `Ctrl+Y`
  - `Alt+Del`
  - `Ctrl+S`
  - `Ctrl+C`

6406534366700. ✖

6406534366701. ✖

- `vi filename.txt` (then in vi command mode):
  - `G1`
  - `x`
  - `:w`, then `:q`
- `nano filename.txt` (then in nano):
  - `Ctrl+B`
  - `Ctrl+D`
  - `Ctrl+S`
  - `Ctrl+Z`

- `vi filename.txt` (then in vi command mode):
  - `1gg`
  - `dd`
  - `:q!`
- `nano filename.txt` (then in nano):
  - `Alt+\`
  - `Ctrl+K`
  - `Ctrl+X` (without saving)

6406534366702. ✖

✔ Official Answer Key: Option (A)

SC_032 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 233 Question Id : 6406531298627 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You have two commands: `verify_data` and `create_report`. You want to execute `create_report` only if `verify_data` completes successfully. If `verify_data` is successful, the output of `create_report` should overwrite a file named `final_report.txt`. If `verify_data` fails, no report should be created.

Which of the following command sequences correctly implements this logic?

Options :

6406534366707. ✖ `verify_data ; create_report > final_report.txt`

6406534366708. ✖ `verify_data || create_report > final_report.txt`

6406534366709. ✔ `verify_data && create_report > final_report.txt`

6406534366710. ✖ `verify_data | create_report > final_report.txt`

✔ Official Answer Key: Option (C)

SC_040 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 241 Question Id : 6406531298636 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

You want to create a new directory named `Backup` inside your `Temp` directory, then copy `report.txt` from `Documents` into this new `Backup` directory. Finally, you want to list the contents of the `Backup` directory. Which set of commands achieves this, assuming you are starting from `/home/user/` ?

Options :

6406534366737. ✔

```
mkdir /home/user/Temp/Backup
cp /home/user/Documents/report.txt /home/user/Temp/Backup/
ls /home/user/Temp/Backup
```

6406534366738. ✖

```
mk dir Temp/Backup
copy Documents/report.txt Temp/Backup/
list Temp/Backup
```

6406534366739. ✔

```
cd Temp
mkdir Backup
cp ../Documents/report.txt Backup/
ls Backup
```

6406534366740. ✖

```
newdir /home/user/Temp/Backup
move /home/user/Documents/report.txt /home/user/Temp/Backup/
dir /home/user/Temp/Backup
```

✔ Official Answer Key: Option (A), Option (C)

SC_051 | [MCQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 252 Question Id : 640653470192 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7

Question Label : Multiple Choice Question

Select the standard input that prints "MATCH" as output.

```
read str
pat='^@.*#'

if [[ "$str" =~  "$pat" ]]; then
    echo MATCH
else
    echo NOT MATCH
fi
```

Options :

6406531562492. ✖ !@#$

6406531562493. ✖ arch

6406531562494. ✔ @#

6406531562495. ✖ #@

Sub-Section Number :	5
Sub-Section Id :	64065367732
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (C)

SC_065 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 248 Question Id : 640653737543 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the script that prints the number of files and directory present in the current working directory, in the format "5,7" given that there 5 files and 7 directories present in the current directory before running the script.

Hint:

- Unary operator `-f` checks if the file exists and is a regular file
- Unary operator `-d` checks if the file exists and is a directory

Options :

6406532468388. ✔

```
#!/bin/bash
files=0
dirs=0
for i in *; do
    if [ -f "$i" ]; then
        files=$((files+1))
    elif [ -d "$i" ]; then
        dirs=$((dirs+1))
    fi
done
echo "$files,$dirs"
```

6406532468389. ✖

```
#!/bin/bash
files=0
dirs=0
for i in *; do
    if [ -f "$i" ]; then
        files=$((files+1))
    elif [ -d "$i" ]; then
        dirs=$((dirs+1))
    fi
done
echo "$dirs,$files"
```

6406532468390. ✖

```
#!/bin/bash
files=0
dirs=0
ls > ls.txt # ls.txt did not exist before
while read -r i; do
    if [ -f "$i" ]; then
        files=$((files+1))
    elif [ -d "$i" ]; then
        dirs=$((dirs+1))
    fi
done < ls.txt
echo "$files,$dirs"
```

6406532468391. ✔

✔ Official Answer Key: Option (A), Option (D)

SC_094 | [MCQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 242 Question Id : 640653577996 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7

Question Label : Multiple Choice Question

The following script takes list of file names as input and finds if the file is executable and counts and lists out executable files. Identify the missing code block from the following options.

```
#!/bin/bash

files=("$@")
executable_files=()

********************
 ***Missing Code***
********************


echo "Number of executable files: ${#executable_files[@]}"
echo "Executable files:"
for file in "${executable_files[@]}"
do
    echo "$file"
done
```

Options :

6406531930139. ✔

```
for file in "${files[@]}"
do
    if [ -x "$file" ]
    then
        executable_files+=("$file")
    fi
done
```

6406531930140. ✘

```
for file in "${files[#]}"
do
    if [ -x "$file" ]
    then
        executable_files+=("$file")
    fi
done
```

6406531930141. ✘

```
for file in "${files[@]}"
do
    if [ -x "$file" ]
    then
        executable_files=("$file") + 1
    fi
done
```

6406531930142. ✘

```
for file in "${files[@]}"
do
    if [ -f "$file" ]
    then
        executable_files+=("$file")
    fi
done
```

✔ Official Answer Key: Option (A)

SC_097 | [MSQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 245 Question Id : 640653577998 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

Following file contains the information on the websites visited on certain server. From the options identify the correct regular expression (ERE) which can capture the website domain names (https://www.something.something or http://www.something.something).

Website	IP Address	Hits	Server Location
https://www.bechtelar.com/alias-similique-ratione-voluptates-aliquam-delectus-qui-cumque-aut	204.83.121.207	0	CK
http://www.okeefe.info/quis-repudiandae-nobis-repellendus-omnis-dolor	78.86.32.75	6	SE
http://macejkovic.com/aut-qui-nostrum-numquam.html	74.58.20.242	1	MU
http://www.cummings.net/magnam-excepturi-eos-rerum-dolores	82.204.55.211	8	PH
http://www.barton.biz/iure-iusto-explicabo-est-soluta-recusandae	252.194.135.149	1	SO
http://www.haag.com/et-exercitationem-id-sunt-sed-laboriosam	1.118.48.149	4	PW
http://www.beer.com/voluptatem-quod-nesciunt-aut.html	71.88.92.193	9	SJ
http://www.mann.org/omnis-ex-in-est-et.html	249.210.50.146	9	BZ
http://towne.com/	58.17.4.75	9	WF
http://www.feeney.net/enim-animi-sapiente-porro-aut-velit-dicta	201.58.189.12	3	ML
http://heaney.com/aliquid-et-rerum-porro-nesciunt-voluptate-quo-sint	9.24.3.149	7	KM
http://bode.com/sint-ut-et-possimus-odit-debitis.html	210.73.162.76	6	BQ

Options :

6406531930147. ✖ `https?://[a-zA-Z]+\.[a-zA-Z]{2,}`

6406531930148. ✔ `http.*//[^/]+`

6406531930149. ✔ `https?://[^/]+`

✔ Official Answer Key: Option (B), Option (C)

SC_108 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 164 Question Id : 640653825167 Question Type : MSQ

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

You try to SSH into a remote VM using the command `ssh user@remote-vm` and receive the following error:

```
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@    WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!     @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
```

Which of the following option(s) are likely the cause?

Options :

6406532775159. ✔ The remote VM's SSH host key has changed.

6406532775160. ✔ The remote VM's IP address has changed and is now associated with a different host key.

6406532775161. ✖ The local `~/.ssh/known_hosts` file is corrupted.

6406532775162. ✖ The remote VM's SSH server is down.

✔ Official Answer Key: Option (A), Option (B)

SC_110 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 166 Question Id : 640653825172 Question Type : MSQ
Correct Marks : 6 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which of the following commands will create a directory `projects` with a subdirectory `2024`, then create a symbolic link named `current` pointing to `2024`, and finally verify the link?

Hint

```
MKDIR(1)                                                    User Commands
MKDIR(1)

NAME
       mkdir - make directories

SYNOPSIS
       mkdir [OPTION]... DIRECTORY...

DESCRIPTION
       Create the DIRECTORY(ies), if they do not already exist.

...
       -p, --parents
              no error if existing, make parent directories as needed
...
LN(1)                                                       User Commands
LN(1)

NAME
       ln - make links between files

SYNOPSIS
       ln [OPTION]... [-T] TARGET LINK_NAME
       ln [OPTION]... TARGET
       ln [OPTION]... TARGET... DIRECTORY
       ln [OPTION]... -t DIRECTORY TARGET...

DESCRIPTION
       In the 1st form, create a link to TARGET with the name LINK_NAME.  In
the 2nd form, create a link to TARGET in the current directory.
       In the 3rd and 4th forms, create links to each TARGET in DIRECTORY.
Create hard links by default, symbolic  links  with  --symbolic.
       By  default,  each  destination (name of new link) should not already
exist.  When creating hard links, each TARGET must exist.  Sym-
       bolic links can hold arbitrary text; if later resolved, a relative link
is interpreted in relation to its parent directory.
```

✔ Official Answer Key: Option (A), Option (B)

SC_112 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 168 Question Id : 640653825173 Question Type : MCQ
Correct Marks : 7
Question Label : Multiple Choice Question

You have a file `sample.txt` and its SHA1 checksum stored in `sample.sha1`. You want to ensure the checksum matches but only print a message if it fails. Which command sequence will achieve this?

```
SHA1SUM(1)                                                    User Commands
SHA1SUM(1)

NAME
       sha1sum - compute and check SHA1 message digest

SYNOPSIS
       sha1sum [OPTION]... [FILE]...

DESCRIPTION
       Print or check SHA1 (160-bit) checksums.

       With no FILE, or when FILE is -, read standard input.

       -b, --binary
              read in binary mode

       -c, --check
              read SHA1 sums from the FILEs and check them

       --tag  create a BSD-style checksum
       -t, --text
              read in text mode (default)

       -z, --zero
              end each output line with NUL, not newline, and disable file
name escaping
...
```

Options :

6406532775183. ✔
```
sha1sum -c sample.sha1 > /dev/null || echo "Checksum does not match"
```

6406532775184. ✘
```
sha1sum sample.txt -c sample.sha1 > /dev/null || echo "Checksum does not
match"
```

6406532775185. ✘
```
sha1sum -c sample.sha1 > /dev/null && echo "Checksum does not match"
```

✔ Official Answer Key: Option (A)

SC_119 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+8 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 131 Question Id : 6406531527865 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You execute the following command:

```
chmod 554 file1.txt
```

Which of the following actions are allowed on `file1.txt`?

Options :

6406535115731. ✖ Owner can only execute, group can only write, others can read

6406535115732. ✔ Owner can read and execute, group can read and execute, others can read

6406535115733. ✖ Owner can read and execute, group can write and execute, others can read

6406535115734. ✖ Owner can read and execute, group can read and write, others can read and write

✔ **Official Answer Key: Option (B)**

SC_122 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 134 Question Id : 6406531527868 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
You have a file named `greetings.txt` with the following content:

```
Hello World
Hello everyone
Say Hello to your friends
```

You want to:

Replace all occurrences of the word `Hello` with `Hi` in the file, then save and exit.

Which of the following sequences correctly performs this in both `vi` and `nano`?

Options :

- `vi greetings.txt`
  - `:s/Hello/Hi/`
  - `:wq`
- `nano greetings.txt`
  - Manually navigate to each `Hello` and type `Hi`
  - `Ctrl+O`, Enter → `Ctrl+X`

6406535115743. ✖

- `vi greetings.txt`
  - `:%s/Hello/Hi/g`
  - `:wq`
- `nano greetings.txt`
  - Use `Ctrl+\` (replace)
  - Enter `Hello` → Enter `Hi` → Press `A` for all
  - `Ctrl+O`, Enter → `Ctrl+X`

6406535115744. ✔

- `vi greetings.txt`
  - `/Hello`, `RHi`
  - `:x`
- `nano greetings.txt`
  - `Ctrl+W`, `Hello`, `Ctrl+D`
  - `Ctrl+O`, `Ctrl+X`

6406535115745. ✖

- `vi greetings.txt`
  - `gg`, `dd`, `iHi`, `Esc`, `:wq`
- `nano greetings.txt`
  - `Ctrl+A`, `Ctrl+K`, type `Hi`, `Ctrl+X`

6406535115746. ✖

✔ Official Answer Key: Option (B)

SC_136 | [MCQ] | Marks: +3.0 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025 (+5 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 124 Question Id : 6406531525137 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

You have a file named `greetings.txt` with the following content:

```
Hello World
Hello everyone
Say Hello to your friends
```

You want to:

Replace all occurrences of the word `Hello` with `Hi` in the file, then save and exit.

Which of the following sequences correctly performs this in both `vi` and `nano`?

Options :

- `vi greetings.txt`
  - `:s/Hello/Hi/`
  - `:wq`
- `nano greetings.txt`
  - Manually navigate to each `Hello` and type `Hi`
  - `Ctrl+O`, Enter → `Ctrl+X`

6406535108106. ✖

- `vi greetings.txt`
  - `:%s/Hello/Hi/g`
  - `:wq`
- `nano greetings.txt`
  - Use `Ctrl+\` (replace)
  - Enter `Hello` → Enter `Hi` → Press `A` for all
  - `Ctrl+O`, Enter → `Ctrl+X`

6406535108107. ✔

- `vi greetings.txt`
  - `/Hello, RHi`
  - `:x`
- `nano greetings.txt`
  - `Ctrl+W, Hello, Ctrl+D`
  - `Ctrl+O, Ctrl+X`

6406535108108. ✖

6406535108109. ✖

✔ Official Answer Key: Option (B)

SC_161 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 138 Question Id : 640653995574 Question Type : MCQ
Correct Marks : 8
Question Label : Multiple Choice Question

You receive a **"Host key verification failed"** error when trying to SSH into a remote VM. Which `ssh-keygen` command would you use to remove the old host key?

Hint
Excerpts from MAN page of `ssh-keygen` command



DESCRIPTION
ssh-keygen generates, manages and converts authentication keys for ssh(1). ssh-keygen can create keys for use by SSH protocol version 2.

The type of key to be generated is specified with the -t option. If invoked without any arguments, ssh-keygen will generate an RSA key.

ssh-keygen is also used to generate groups for use in Diffie-Hellman group exchange (DH-GEX). See the MODULI GENERATION section for details.

Finally, ssh-keygen can be used to generate and update Key Revocation Lists, and to test whether given keys have been revoked by one. See the KEY REVOCATION LISTS section for details.

Normally each user wishing to use SSH with public key authentication runs this once to create the authentication key in ~/.ssh/id_dsa, ~/.ssh/id_ecdsa, ~/.ssh/id_ecdsa_sk, ~/.ssh/id_ed25519, ~/.ssh/id_ed25519_sk or ~/.ssh/id_rsa. Additionally, the system administrator may use this to generate host keys.

Normally this program generates the key and asks for a file in which to store the private key. The public key is stored in a file with the same name but ".pub" appended. The program also asks for a passphrase. The passphrase may be empty to indicate no passphrase (host keys must have an empty passphrase), or it may be a string of arbitrary length. A passphrase is similar to a password, except it can be a phrase with a series of words, punctuation, numbers, whitespace, or any string of characters you want. Good passphrases are 10-30 characters long, are not simple sentences or otherwise easily guessable (English prose has only 1-2 bits of entropy per character, and provides very bad passphrases), and contain a mix of upper and lowercase letters, numbers, and non-alphanumeric characters. The passphrase can be changed later by using the -p option.

There is no way to recover a lost passphrase. If the passphrase is lost or forgotten, a new key must be generated and the corresponding public key copied to other machines.

ssh-keygen will by default write keys in an OpenSSH-specific format. This format is preferred as it offers better protection for keys at rest as well as allowing storage of key comments within the private key file itself. The key comment may be useful to help identify the key. The comment is initialized to "user@host" when the key is created, but can be changed using the -c option.

It is still possible for ssh-keygen to write the previously-used PEM format private keys using the -m flag. This may be used when generating new keys, and existing new-format keys may be converted using this option in conjunction with the -p (change passphrase) flag.

After a key is generated, ssh-keygen will ask where the keys should be placed to be activated.

...
-t dsa | ecdsa | ecdsa-sk | ed25519 | ed25519-sk | rsa
Specifies the type of key to create. The possible values are "dsa", "ecdsa", "ecdsa-sk", "ed25519", "ed25519-sk", or "rsa".
-b bits
Specifies the number of bits in the key to create. For RSA keys, the minimum size is 1024 bits and the default is 3072 bits. Generally, 3072 bits is considered sufficient. DSA keys must be exactly 1024 bits as specified by FIPS 186-2. For ECDSA keys, the -b flag determines the key length by selecting from one of three elliptic curve sizes: 256, 384 or 521 bits. Attempting to use bit lengths other than these three values for ECDSA keys will fail. ECDSA-SK, Ed25519 and Ed25519-SK keys have a fixed length and the -b flag will be ignored.
...
-f filename
Specifies the filename of the key file.

-R hostname | [hostname]:port
Removes all keys belonging to the specified hostname (with optional port number) from a known_hosts file. This option is useful to delete hashed hosts (see the -H option above).
...

Options :

6406533361042. ✔ `ssh-keygen -R remote-vm`

6406533361043. ✖ `ssh-keygen -t rsa -b 4096`

6406533361044. ✖ `ssh-keygen -f ~/.ssh/id_rsa -y`

6406533361045. ✖ `ssh-keygen -L remote-vm`

✔ Official Answer Key: Option (A)

SC_165 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 142 Question Id : 640653995572 Question Type : MSQ
Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

You try to SSH into a remote VM using the command `ssh user@remote-vm` and receive the following error:

```
Host key verification failed.
```

What are the possible reasons for this error?

Options :

6406533361034. ✔ The remote VM's IP address has changed.

6406533361035. ✔ The remote VM's SSH host key has changed.

6406533361036. ✖ The local SSH configuration file is corrupted.

6406533361037. ✖ The SSH client is not installed.

✔ Official Answer Key: Option (A), Option (B)

SC_166 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 143 Question Id : 640653995573 Question Type : MSQ

Correct Marks : 8 Max. Selectable Options : 0

Question Label : Multiple Select Question

You try to SSH into a remote VM using the command `ssh user@remote-vm` and receive the following error:

```
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@    WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!     @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
```

Which of the following option(s) are likely the cause ?

Options :

6406533361038. ✔ The remote VM's SSH host key has changed.

6406533361039. ✔ The remote VM's IP address has changed and is now associated with a different host key.

6406533361040. ✖ The local `~/.ssh/known_hosts` file is corrupted.

6406533361041. ✖ The remote VM's SSH server is down.

Sub-Section Number :	6
Sub-Section Id :	640653146824
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A), Option (B)

SC_169 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 146 Question Id : 640653995579 Question Type : MSQ
Correct Marks : 6 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which of the following command sequences will create a SHA256 checksum for the file `data.bin` and later verify its integrity?

```
SHA256SUM(1)                                                    User
Commands                                                     SHA256SUM(1)

NAME
       sha256sum - compute and check SHA256 message digest

SYNOPSIS
       sha256sum [OPTION]... [FILE]...
DESCRIPTION
       Print or check SHA256 (256-bit) checksums.

       With no FILE, or when FILE is -, read standard input.

       -b, --binary
              read in binary mode

       -c, --check
              read SHA256 sums from the FILEs and check them

       --tag  create a BSD-style checksum

       -t, --text
              read in text mode (default)

       -z, --zero
              end each output line with NUL, not newline, and disable file
name escaping

...
```

Options :

6406533361062. ✔
```
sha256sum data.bin > data.sha256
sha256sum -c data.sha256
```

6406533361063. ✖
```
sha256sum data.bin -o data.sha256
sha256sum -v data.sha256
```

6406533361064. ✔

✔ Official Answer Key: Option (A), Option (C)

SC_174 | [SA] | Marks: +3.0 | *System commands*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 2 Question Id : 6406532159052 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

A Linux-like file system uses **index nodes (inodes)** to manage files. A newly formatted volume has 10,000 free inodes. A user performs the following sequential actions:

1. A directory named `/projects` is created.
2. Inside `/projects`, a script creates a single large data file named `dataset.bin`.
3. The script then successfully creates $1,500$ **hard links** to `dataset.bin` within the same directory, each with a unique name.
4. Finally, the script successfully creates 500 **symbolic (soft) links**, each pointing to `dataset.bin`.

Assume each symbolic link occupies its own inode and hard links do not allocate new inodes.

Assuming no other processes are writing to the volume, what is the total number of **free inodes** remaining on the volume after these actions are completed?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

9498

Sub-Section Number :	3
Sub-Section Id :	640653396305
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (Numerical Value): 9498**

SC_178 | [MCQ] | Marks: +3.0 | *System commands*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 6 Question Id : 6406532159054 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
A system directory contains the following list of files:

- `backup_2025.tar`
- `backup_2026.log`
- `backup_26.tar`
- `backup_v2.tar`
- `backup_a.log`

An administrator wants to process a specific subset of these files using the following command:

```
ls backup_[0-9][0-9]*.tar
```

Which of the following statements correctly describes which files are matched by this command and why?

Options :

6406537023164. ✖ It matches only `backup_26.tar` because the `*` wildcard acts as a quantifier requiring exactly two digits total.

6406537023165. ✔ It matches `backup_2025.tar` and `backup_26.tar` because the `*` wildcard matches any sequence of characters (including empty strings) following the first two digits.

6406537023166. ✖ It matches `backup_2025.tar`, `backup_2026.log`, and `backup_26.tar` because wildcards match files based only on the prefix before the first bracket.

6406537023167. ✖ It matches `backup_2025.tar`, `backup_26.tar`, and `backup_v2.tar` because `[0-9]` matches any single alphanumeric character.

✔ Official Answer Key: Option (B)

SC_181 | [MCQ] | Marks: +3.0 | *System commands*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 9 Question Id : 6406532159057 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A system administrator writes a maintenance script named `clean_logs.sh`. They attempt to run the script with the following redirection setup in the terminal:

```
./clean_logs.sh 2> debug.log > debug.log
```

What is the operational outcome of executing this specific command?

Options :

6406537023176. ✖ Both standard output (`stdout`) and standard error (`stderr`) are appended to `debug.log` because both redirections use the same filename.

6406537023177. ✖ The command results in a syntax error because the same file cannot be used for multiple redirections in a single command.

6406537023178. ✔ Both `stdout` and `stderr` are redirected to `debug.log`, but the file is opened independently for each file descriptor. As a result, output from the two streams may interleave, overwrite portions of each other, or appear in an unpredictable order.

6406537023179. ✖ The shell detects that both streams target the same file and automatically treats the command as `> debug.log 2>&1`.

✔ Official Answer Key: Option (C)

SC_182 | [MCQ] | Marks: +3.0 | *System commands*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.2 Shell Scripting: Conditionals & Test Operators

Question Number : 10 Question Id : 6406532159058 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You want to perform an automated network diagnostics sequence with the following constraints:

1. Run a primary connection test script `./ping_server.sh`.
2. **If and only if it fails**, run an intensive route tracing script `./trace_route.sh`.
3. From the output of `./trace_route.sh`, extract **only** the lines containing the word `DROP`.
4. Save those filtered lines to a file named `network_errors.log`.

Which command combination is correct?

Options :

6406537023180. ✔ `./ping_server.sh || { ./trace_route.sh | grep "DROP" > network_errors.log; }`

6406537023181. ✖ `./ping_server.sh && { ./trace_route.sh | grep "DROP" > network_errors.log; }`

6406537023182. ✖ `./ping_server.sh || { ./trace_route.sh > network_errors.log; }`

6406537023183. ✖ `./ping_server.sh || { ./trace_route.sh | grep "DROP"; }`

✔ **Official Answer Key: Option (A)**

• 4.4 Process Management & Signals **(3 Questions)**

SC_012 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.4 Process Management & Signals

Question Number : 265 Question Id : 640653345413 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Select Question

Consider the file hierarchy structure given below to answer the question. All the file names have an extension e.g. `.sh`, `.txt` etc, and the rest are directories. The top-most directory is `/home`.

```
/home
├── classA
│   └── attendance
│       ├── feb.csv
│       ├── jan.csv
│       └── mar.csv
├── classB
│   └── attendance
│       ├── feb.csv
│       ├── jan.csv
│       └── mar.csv
├── classC
│   └── attendance
│       ├── feb.csv
│       ├── jan.csv
│       └── mar.csv
├── aprClassB.csv
└── script.sh

6 directories, 10 files
```

Which of the following commands can be used to rename the file `aprClassB.csv` to `apr.csv` and move it to the directory `attendance` in the directory `classB`. Consider that the current working directory is `/home/classB`.

Options :

6406531148822. ✗ `cp aprClassB.csv apr.csv`

6406531148823. ✗ `cp ../aprClassB.csv attendance/apr.csv`

6406531148824. ✓ `mv ../aprClassB.csv attendance/apr.csv`

6406531148825. ✓ `mv /home/aprClassB.csv /home/classB/attendance/apr.csv`

✔ Official Answer Key: Option (C), Option (D)

SC_111 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.4 Process Management & Signals

Question Number : 167 Question Id : 640653825163 Question Type : MCQ

Correct Marks : 7

Question Label : Multiple Choice Question

Which of the following commands will show its process ID (PID) and run in the background?

Options :

6406532775146. ✔ `command & echo $!`

6406532775147. ✖ `command && echo $!`

6406532775148. ✖ `command &; echo $!`

6406532775149. ✖ `command &`

✔ **Official Answer Key: Option (A)**

SC_183 | [MCQ] | Marks: +3.0 | *System commands*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.4 Process Management & Signals

Question Number : 11 Question Id : 6406532159062 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following commands will print lines 8 to 15 (inclusive) from the top of a file named `system.log`?

Options :

6406537023196. ✔ `cat system.log | head -15 | tail -8`

6406537023197. ✖ `cat system.log | tail -15 | head -8`

6406537023198. ✖ `cat system.log | head -8 | tail -15`

6406537023199. ✖ `cat system.log | tail -8 | head -15`

✔ **Official Answer Key: Option (A)**

• 4.3 Shell Scripting: Loops, Functions & Positional Parameters (10 Questions)

SC_026 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.3 Shell Scripting: Loops, Functions & Positional Parameters

Question Number : 227 Question Id : 6406531298621 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the fundamental difference between a hard link and a symbolic link. Which statement best describes this difference?

Options :

6406534366683. ✖ A hard link contains a copy of the original file's data, while a symbolic link does not.

6406534366684. ✔ A hard link is essentially another name for the same file data (inode), while a symbolic link is a separate file containing the path to the original file.

6406534366685. ✖ Hard links can cross filesystem boundaries, whereas symbolic links cannot.

6406534366686. ✖ Hard links are exclusively used for pointing to directories, while symbolic links can only point to regular files.

✔ Official Answer Key: Option (B)

SC_055 | [MSQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.3 Shell Scripting: Loops, Functions & Positional Parameters

Question Number : 256 Question Id : 640653470193 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 9 Selectable Option : 0

Question Label : Multiple Select Question

The files inside the directory is given below.

```
-rw-rw-r--  1 meena meena        2 Nov 30 09:53  docker_notes.txt
-rwxrwxr-x  1 meena meena     2901 Nov 18 13:47  emacs_notes.txt
drwxrwxr-x 11 meena meena     4096 Jul 27 17:24  vim
-rw-rw-r--  1 meena meena     1709 Nov 18 13:47  vim_notes.txt
drw-rw-r--  1 meena meena     4096 Nov 18 13:47  x
drw-rw-r--  1 meena meena     4096 Nov 18 13:47  y
drw-rw-r--  1 meena meena     4096 Nov 18 13:47  z
```

Select the file(s) that will be printed in the output. [MSQ]

```
for file in *; do
    # -f returns exit status 0 when the operand is a file
    [ -f $file ] && echo $file
done
```

Options :

6406531562496. ✔ docker_notes.txt

6406531562497. ✔ emacs_notes.txt

6406531562498. ✔ vim_notes.txt

6406531562499. ✖ x

6406531562500. ✖ y

6406531562501. ✖ z

Sub-Section Number :	8
Sub-Section Id :	64065367735
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A), Option (B), Option (C)

SC_056 | [MSQ] | Marks: +3.0 | *Quiz 1 Jan23*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.3 Shell Scripting: Loops, Functions & Positional Parameters

Question Number : 257 Question Id : 640653470194 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6 Selectable Option : 0

Question Label : Multiple Select Question

```
while read line; do
    echo $line
done < pattern.txt
```

Select the equivalent command for the above script.

Options :

6406531562502. ✔ `cat pattern.txt`

6406531562503. ✖ `cat pattern.txt | cat > pattern.txt`

6406531562504. ✖ `cat > pattern.txt`

6406531562505. ✔ `grep . pattern.txt`

✔ Official Answer Key: Option (A), Option (D)

SC_072 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.3 Shell Scripting: Loops, Functions & Positional Parameters

Question Number : 255 Question Id : 640653737542 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Short Answer Question

Assuming the file `myfile.txt` has 10 lines, what will be the output from the following script?

```
#!/bin/bash
c=0
while read -r line; do
        c=$((c+1))
        if [ "$c" -eq 5 ]; then
                break
        fi
done < myfile.txt
echo $c
```

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

✔ Official Answer Key (Numerical Value): 5

SC_076 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.3 Shell Scripting: Loops, Functions & Positional Parameters

Question Number : 244 Question Id : 6406531113963 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider that a bash script is written in a file named `script.sh` in the current working directory.

Which of the following commands will correctly set the permissions to allow the owner and group to read, write and execute the script, while denying all permissions to others?

Options :

6406533774706. ✖ `chmod 600 script.sh`

6406533774707. ✔ `chmod 770 script.sh`

6406533774708. ✖ `chmod 740 script.sh`

6406533774709. ✖ `chmod 750 script.sh`

✔ **Official Answer Key: Option (B)**

SC_104 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.3 Shell Scripting: Loops, Functions & Positional Parameters

Question Number : 160 Question Id : 640653825168 Question Type : MCQ
Correct Marks : 6
Question Label : Multiple Choice Question

You encounter a "Permission denied (publickey)" error when trying to SSH into a remote VM. Which `ssh-keygen` command would you use to ensure your SSH key has the correct permissions?

Hint
Excerpts from MAN page of `ssh-keygen` command

```
DESCRIPTION
     ssh-keygen generates, manages and converts authentication keys for
ssh(1).  ssh-keygen can create keys for use by SSH protocol version
     2.

     The type of key to be generated is specified with the -t option.  If
invoked without any arguments, ssh-keygen will generate an RSA
     key.

     ssh-keygen is also used to generate groups for use in Diffie-Hellman
group exchange (DH-GEX).  See the MODULI GENERATION section for
     details.

     Finally, ssh-keygen can be used to generate and update Key Revocation
Lists, and to test whether given keys have been revoked by one.
     See the KEY REVOCATION LISTS section for details.

     Normally each user wishing to use SSH with public key authentication runs
this once to create the authentication key in ~/.ssh/id_dsa,
     ~/.ssh/id_ecdsa, ~/.ssh/id_ecdsa_sk, ~/.ssh/id_ed25519,
~/.ssh/id_ed25519_sk or ~/.ssh/id_rsa.  Additionally, the system administrator
     may use this to generate host keys.

     Normally this program generates the key and asks for a file in which to
store the private key.  The public key is stored in a file with
     the same name but ".pub" appended.  The program also asks for a
passphrase.  The passphrase may be empty to indicate no passphrase
     (host keys must have an empty passphrase), or it may be a string of
arbitrary length.  A passphrase is similar to a password, except it
     can be a phrase with a series of words, punctuation, numbers, whitespace,
or any string of characters you want.  Good passphrases are
     10-30 characters long, are not simple sentences or otherwise easily
guessable (English prose has only 1-2 bits of entropy per charac-
     ter, and provides very bad passphrases), and contain a mix of upper and
lowercase letters, numbers, and non-alphanumeric characters.
```

✔ Official Answer Key: Option (B)

SC_127 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025 (+8 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.3 Shell Scripting: Loops, Functions & Positional Parameters

Question Number : 139 Question Id : 6406531527874 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

You want to:

- Try to extract the **first 5 lines** of `input.txt`
- If successful, write "Done" to `status.log`
- If it fails, write "Error in input" to `error.log`

Which command does this?

Options :

6406535115767. ✔ `head -n 5 input.txt && echo "Done" > status.log || echo "Error in input" > error.log`

6406535115768. ✖ `head -n 5 input.txt || echo "Done" > status.log && echo "Error in input" > error.log`

6406535115769. ✖ `head -n 5 input.txt | echo "Done" > status.log || echo "Error in input" > error.log`

6406535115770. ✖ `head -n 5 input.txt && echo "Error in input" > error.log || echo "Done" > status.log`

Sub-Section Number :	4
Sub-Section Id :	640653244932
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

SC_148 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.3 Shell Scripting: Loops, Functions & Positional Parameters

Question Number : 228 Question Id : 640653668674 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **TRUE** statements for hardlinks and softlinks?

Options :

6406532240227. ✔ Hardlink of a file has same inode number

6406532240228. ✖ Softlink of a file has same inode number

6406532240229. ✖ Softlink of a file occupies the same space (i.e. same size) as the original file

6406532240230. ✔ Hardinks cannot be moved to a different File system

6406532240231. ✖ Softlink of a file unconditionally requires full path

6406532240232. ✔ Changes done in the original file is reflected in its hardlink

Sub-Section Number :	4
Sub-Section Id :	64065395215
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A), Option (D), Option (F)

SC_149 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.3 Shell Scripting: Loops, Functions & Positional Parameters

Question Number : 229 Question Id : 640653668683 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 7 Max. Selectable Options : 0

Question Label : Multiple Select Question

For the following `while` loop, select the standard input(s) from the following options that breaks the loop.

```
while read line; do
    $pat = '[0-3][a-g][123][xyz]'
    [[ $line =~ $pat ]] && break
done
```

Options :

6406532240251. ✖ 8iadaz

6406532240252. ✔ 3g2z

6406532240253. ✖ jkjkjkjk2j2yjkjk

6406532240254. ✖ .a.a.a.2.2.1.

6406532240255. ✔ 00210132a1z01231

6406532240256. ✖ 1a2w01231

Sub-Section Number :	5
Sub-Section Id :	64065395216
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (B), Option (E)

SC_153 | [SA] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.3 Shell Scripting: Loops, Functions & Positional Parameters

Question Number : 233 Question Id : 640653668675 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Short Answer Question

The following commands are executed in the given order. Enter the duration (in number of seconds) after which the echo command will be executed.

```
$ sleep 1
$ $(sleep 10)
$ sleep 100 &
$ echo "Done"
```

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

11

✔ **Official Answer Key (Numerical Value): 11**

- **4.1 Shell Variables, Environment & Quoting** **(3 Questions)**

SC_058 | [MSQ] | Marks: +3.0 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.1 Shell Variables, Environment & Quoting

Question Number : 241 Question Id : 640653737528 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

```
export LOG_FILE="/var/log/$USER.log"
```

From the following options which will print the Shell environmental variable "LOG_FILE"

Options :

6406532468333. ✔ `echo $LOG_FILE`

6406532468334. ✖ `printenv $LOG_FILE`

6406532468335. ✔ `printenv LOG_FILE`

6406532468336. ✖ `env $LOG_FILE`

✔ Official Answer Key: Option (A), Option (C)

SC_077 | [MCQ] | Marks: +3.0 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025 (+2 more papers)*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.1 Shell Variables, Environment & Quoting

Question Number : 245 Question Id : 6406531113964 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following shell variables can be used to get the return code of the last command?

Options :

6406533774710. ✖ `$!`

6406533774711. ✔ `$?`

6406533774712. ✖ `$1`

6406533774713. ✖ `$0`

Sub-Section Number :	3
Sub-Section Id :	640653168577
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

SC_089 | [MCQ] | Marks: +3.0 | *Quiz1 May23*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.1 Shell Variables, Environment & Quoting

Question Number : 237 Question Id : 640653577994 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 6

Question Label : Multiple Choice Question

```
#!/bin/bash
export MY_VAR="Sunset"
my_var="Sunrise"
echo "Var11 is: $MY_VAR"
echo "Var12 is: $my_var"
(
        my_var="Sun_rise"
        MY_VAR_var="Sun_set"
        echo "Var21 is: $MY_VAR"
        echo "Var22 is: $my_var"
        (
                MY_VAR="Sun@rise"
                my_var="Sun@set"
                echo "Var31 is: $MY_VAR"
                echo "Var32 is:${my_var}"
        ) & disown
)
wait
echo "Var41 is: $MY_VAR"
echo "var42 is: $my_var"
```

What will be the value of `Var42` at the end of execution?
Use the following information if needed.

✔ Official Answer Key: Option (A)

• 4.5 Task Scheduling, Archiving & Compression (1 Questions)

SC_175 | [MSQ] | Marks: +3.0 | *System commands*

Category: Week 4: Shell Scripting, Environment Variables & Process Management › 4.5 Task Scheduling, Archiving & Compression

Question Number : 3 Question Id : 6406532159053 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

A system administrator is configuring a shared project directory. The requirements for three specific system objects are defined as follows:

1. `archive.tar` : The owner can only read and write to it. Members of the group can only read it. Everyone else has no access.
2. `deploy.sh` : Anyone on the system can read and execute it, but only the owner can modify it.
3. `logs` : It is a directory. The owner and group members have read, write, and execute permissions on it. External users ("others") have execute permission only, allowing them to access known files without listing the directory contents.

Which of the following lines matches the correct `ls -l` output for these objects?

Options :

6406537023159. ✔ `-rw-r----- 1 admin tech 2048 archive.tar`

6406537023160. ✔ `-rwxr-xr-x 1 admin tech 4096 deploy.sh`

6406537023161. ✖ `drwxrwxr-x 2 admin tech 4096 logs`

6406537023162. ✔ `drwxrwx--x 2 admin tech 4096 logs`

6406537023163. ✖ `-rw-r--r-- 1 admin tech 2048 archive.tar`

✔ Official Answer Key: Option (A), Option (B), Option (D)